The Strong Secretary Conjecture is True for Linear Matroids

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Abstract

We prove a $1/e$ guarantee for Matroid Secretary on linear matroids: each element of a fixed optimal basis is selected with probability at least $1/e$. This settles the strong secretary conjecture for this class of matroids. Our result holds both when the matroid is known in advance and when a linear representation is given online. In the known-matroid model, the result extends to fully modular extendable matroids.

We also obtain $1/e$ guarantees for a more general model on linear matroids, both when the matroid is known and when its representation is revealed online. In XOS Matroid Secretary, agents with XOS valuations over the elements arrive in uniformly random order, and the algorithm must assign each arriving agent a bundle of elements. The bundles must be pairwise disjoint, and their union must be an independent set of the matroid. We obtain an expected total value of at least $1/e$ times the offline optimum. This model contains Group Matroid Secretary, in which the elements are adversarially partitioned into groups that arrive in uniformly random order, and the algorithm selects a subset of each arriving group. For this special case, we select each element of a fixed optimal basis with probability at least $1/e$.

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1 Introduction

The *Matroid Secretary Problem*, introduced by Babaioff, Immorlica, and Kleinberg [5], extends the classical secretary problem from selecting a single element to selecting an independent set of a matroid. Each element e has a fixed nonnegative weight $w(e)$, and the elements arrive in uniformly random order, with weights revealed upon arrival. The algorithm must irrevocably decide whether to accept the element while keeping the selected set independent. Let E denote the ground set. We fix a total order on E before the arrivals and use it to break weight ties in every greedy computation. For every $Y \subseteq E$, let $\mathrm{OPT}(Y)$ denote the resulting greedy maximum-weight basis of the matroid restricted to Y. We say that an algorithm is *c-probability-competitive* if

$$\mathbb{P}\left[e \in \mathrm{ALG}\right] \geq c \ \text{ for every } e \in \mathrm{OPT}(E),$$

where the probability is over the uniformly random arrival order and the internal randomness of the algorithm. For nonnegative weights, this implies an expected weight of at least c times $w(\mathrm{OPT}(E))$. In the classical secretary problem, which corresponds to a rank-one matroid, the best probability guarantee that holds for every number of arrivals is $1/e$.

A matroid M on ground set E is *linear* if there is a field $\mathbb{F}$ and an indexed family of vectors $(v_e)_{e\in E}$ over $\mathbb{F}$ such that $I \subseteq E$ is independent in M if and only if $(v_e)_{e\in I}$ is linearly independent. We call this family a *representation* of M over $\mathbb{F}$. We prove that the optimal $1/e$-probability-competitive guarantee is achievable in the *online-representation model*, in which a representation of the matroid over a finite field is revealed online. The algorithm knows the number of elements and the underlying finite field in advance, but not the representation. Upon arrival, each element reveals its weight and its vector, expressed in a common coordinate system over that field. Thus, the algorithm learns the representation one vector at a time as the elements arrive. In particular, the algorithm does not know the matroid, or a representation of it, before the arrival process begins. The algorithm is ordinal: it uses the weights only through relative comparisons, with ties broken by the fixed total order. We place no restriction on the computation performed by the algorithm. Upon each arrival, it may use any function of the information revealed so far. In particular, we do not bound its running time.

Theorem 1. *In the online-representation model over a finite field, there is an ordinal online algorithm that always selects an independent set* ALG *and satisfies*

$$\mathbb{P}\left[e \in \mathrm{ALG}\right] \geq 1/e \ \ \textit{for every } e \in \mathrm{OPT}(E).$$

Consequently, $\mathbb{E}[w(\mathrm{ALG})] \geq w(\mathrm{OPT}(E))/e$.

Theorem 1 also holds when the vectors are revealed online over an arbitrary field (see Remark 1 at the end of Section 3). The same guarantee is achievable for every linear matroid in the *known-matroid model*. Here the entire matroid is known before the arrival process, while only the weights are revealed online. By a theorem of Rado [45], every finite matroid representable over some field also admits a representation over a finite field. Therefore, we may compute such a representation and apply Theorem 1. In particular, no representation need be supplied as part of the input, and the result applies to matroids representable over an arbitrary field.

The main idea behind our algorithm is to control how much of each subspace is spanned by the accepted vectors. The difficulty is that accepting an element affects all later elements at once: after e is accepted, every element of $\langle \mathrm{ALG}\rangle + \langle e\rangle$ becomes unavailable, where $\langle \mathrm{ALG}\rangle$ denotes the linear span of the vectors selected by the algorithm. It is therefore not enough to control, for each

element, the probability that it can still be accepted. Instead, for every subspace L, we maintain an upper bound on the expected dimension of $L \cap \langle \mathrm{ALG} \rangle$, which we call the *dimension invariant*.

We then choose acceptance probabilities that preserve the dimension invariant while ensuring that an arriving element is accepted with a prescribed probability whenever it belongs to $\mathrm{OPT}(X)$, where X is the set of elements revealed so far. The probability is over both the preceding arrival order and the internal randomness of the algorithm. The above requirements placed on acceptance probabilities lead to a large linear program (LP), and the main technical step in our proof is to establish its feasibility. Duality and uncrossing reduce the feasibility argument to chains of subspaces, for each of which we give an explicit feasible solution. The uncrossing step uses that, for every fixed subspace Z, the dimension of $Z \cap L$ is a supermodular function of the subspace L. This follows from the fact that dimension is modular on subspaces: $\dim(A) + \dim(B) = \dim(A \cap B) + \dim(A+B)$. The rank function of a general matroid is only submodular, meaning this identity holds only as an inequality, which is why our argument relies on a linear representation. More generally, the same argument applies to known matroids that are restrictions of finitary modular matroids, whose lattices of flats satisfy the modular rank identity. In particular, it applies to the fully modular extendable matroids studied by Bérczi, Gehér, Imolay, Lovász, Padró and Schwarcz [9]. This class includes all matroids of rank at most 3 and all skew-representable matroids, that is, matroids representable over a division ring.

The same approach extends to a more general arrival model on linear matroids. In XOS Matroid Secretary, agents with XOS valuations arrive in uniformly random order and receive disjoint bundles whose union is independent. We obtain an expected total value of at least $1/e$ times the offline optimum (Theorem 2 in Section 5). This model contains Group Matroid Secretary, in which the elements are partitioned into groups that arrive in uniformly random order. For this special case, we obtain the stronger guarantee that each element of a fixed greedy optimum is selected with probability at least $1/e$. The extension to agent arrivals follows a similar blueprint to our main result. We select a subset of the bundle assigned to the arriving agent in an optimal allocation of the observed agents. The dimension invariant, together with a polymatroid argument, allows us to choose a feasible random subset with prescribed element-wise marginals. We use an LP to impose these marginals and preserve the invariant. Duality and uncrossing again reduce its feasibility to chains of subspaces. For each chain, we add auxiliary vectors to the bundle before selecting the random subset, then discard those vectors. For XOS valuations, these marginals give a corresponding fraction of the bundle's value in expectation. For groups, each element of the global greedy optimum also belongs to the greedy optimum of the observed elements when its group arrives. We can therefore use the same marginals to bound its selection probability.

We first present our main result for the Matroid Secretary Problem in Sections 2 and 3. In Section 4, we show how the same guarantee extends to fully modular extendable matroids. We then develop the XOS and group arrival case independently in Section 5, replicating the shared steps so that the proof is self-contained.

1.1 Related Work

The classical secretary problem was studied by Lindley [41] and Dynkin [25]. Kleinberg [37] studied the multiple-choice secretary problem, in which up to k elements may be selected. Later, Babaioff, Immorlica, and Kleinberg [5] introduced the Matroid Secretary Problem; see also the journal version with Kempe [4]. They conjectured that a constant expected-weight (or utility-competitive) guarantee is achievable for every matroid.

For general rank-r matroids, Babaioff, Immorlica, and Kleinberg [5] and Babaioff, Immorlica, Kempe, and Kleinberg [4] obtained an $\Omega(1/\log r)$ expected-weight guarantee. This was improved to

$\Omega(1/\sqrt{\log r})$ by Chakraborty and Lachish [12], and then to $\Omega(1/\log\log r)$ by Lachish [39]. Feldman, Svensson, and Zenklusen [28] later gave a simpler algorithm achieving the same guarantee.

A stronger notion is probability-competitiveness, which requires a guarantee for each element of the optimum. Bateni, Hajiaghayi, and Zadimoghaddam [8] obtained an $\Omega(1/\log^2 r)$ probability-competitive guarantee, and Soto, Turkieltaub, and Verdugo [51] improved it to $\Omega(1/\log r)$. The strong Matroid Secretary Conjecture asks whether the optimal rank-one guarantee $1/e$ can be achieved under this criterion for every matroid [51]. For linear matroids, we obtain the $1/e$ probability-competitive guarantee in the known-matroid and online-representation models. Constant guarantees were previously known for transversal [18, 35, 51], graphic [3, 5, 7, 10, 38, 51], laminar [10, 32, 33, 34, 42, 51], and regular matroids [19].

Recent and concurrent results. Several results on the Matroid Secretary Problem appeared in rapid succession over a few weeks. Singla [49] resolved the original conjecture with an ordinal 1/4-probability-competitive algorithm for arbitrary matroids in the oracle model, requiring only n in advance. His approach reduces the problem to a matroid version of the random-order two-sided Game of Googol [16] via reversible updates that preserve greedy decisions on finalized elements; using the reduction framework of Feldman, Svensson, and Zenklusen [29], he also obtained an $\Omega(1/k^3)$-competitive algorithm for the intersection of k matroids. Building on Singla's approach, Chan, Lin, and Wang [13] improved the selection probability to $1/c \approx 1/3.1462$, where $c > 1$ satisfies $c - \ln c = 2$, via random deletions and time-dependent acceptance.

Independently and concurrently, Abdi, Banihashem, Hajiaghayi, and Mittal [1] obtained an ordinal $1/e$-probability-competitive algorithm for known linear matroids using linear programming, with a feasibility proof based on the same subspace dimension invariant and uncrossing argument as ours. They also proved a tight 1/2 single-sample prophet inequality for arbitrary matroids, yielding a 1/64-competitive secretary algorithm in the oracle model.

Finally, Huang [31] settled the strong Matroid Secretary Conjecture for arbitrary matroids with an ordinal, expected-polynomial-time $1/e$-probability-competitive algorithm in the oracle model, matching the optimal rank-one bound via a continuous-time formulation.

The algorithms of Singla and Huang apply to arbitrary matroids using only independence-oracle access, but they process one element at a time. They do not directly apply when elements arrive in groups or are allocated to agents. In Group Matroid Secretary, the groups arrive in random order, but the elements do not: the elements of a group arrive together. In XOS Matroid Secretary, an arriving agent receives a bundle, so several elements must be selected jointly. Our dimension bounds hold for every subspace, not only for the lines spanned by single elements. Together with a polymatroid argument, they allow us to select a random subset of a bundle with prescribed marginals while preserving the invariant. We thus obtain $1/e$ guarantees for both models on linear matroids. These guarantees are optimal, since both models contain the classical secretary problem. To our knowledge, no constant-competitive algorithms were previously known for these models on general linear matroids.

Linear-programming approaches. Linear programming has been previously used to design and analyze secretary algorithms. Buchbinder, Jain, and Singh [11] gave a finite LP characterization of secretary algorithms, and Chan, Chen, and Jiang [14] developed a continuous LP approach for generalized secretary problems. Related LP formulations have since been used in secretary problems with advice and in sample-driven optimal stopping [15, 23].

Matroid information model. The information available about the matroid gives rise to two standard variants [17, 44]. In the *known-matroid model*, the ground set and the entire independence structure are available before the arrival process, and only the weights are revealed online.

In the *unknown-matroid model*, the algorithm knows the number of elements in advance but can query independence only for subsets of elements that have already arrived. In particular, it has no information about dependencies involving unseen elements. In the online-representation model considered in this paper, the representation is not known in advance, but each arriving element reveals its vector in a common coordinate system. From these vectors, we can answer every independence query on observed elements. We can also construct subspaces of the observed span and choose auxiliary vectors in them. The subspaces need not be spanned by observed elements, and the auxiliary vectors need not correspond to elements of the matroid.

For many special classes of matroids, the natural online model similarly reveals a representation of each arriving element. For graphic matroids, an arriving edge reveals its endpoints, while for transversal matroids an arriving element reveals its neighborhood in the underlying bipartite representation. This additional information has been important in obtaining constant guarantees. For example, the $1/e$ guarantee of Kesselheim, Radke, Tönnis, and Vöcking [35] for transversal matroids uses this bipartite representation. Similarly, the strongest previous guarantees for graphic matroids use the graph representation [7, 10], whereas Dütting, Paes Leme, Pál, and Patel [24] recently obtained a $1/36$ utility-competitive guarantee using only independence queries on arrived elements. Our online-representation model takes the analogous viewpoint for linear matroids.

Cristi, Dütting, Kleinberg, Paes Leme, and Patel [17] introduced online matroid embeddings, which map a matroid revealed online into a larger known matroid. They constructed embeddings for binary and laminar matroids and obtained reductions from unknown to known matroids with an arbitrarily small loss in the expected-weight guarantee. They also proved limitations of this embedding approach for general matroids.

Other approaches and models. Dughmi [20, 21] used duality to establish an equivalence between constant-competitive matroid secretary algorithms and random-order contention resolution for possibly correlated inputs. On the negative side, Bahrani, Beyhaghi, Singla, and Weinberg [6] proved limitations of broad classes of greedy and partition-based algorithms. Abdolazimi, Karlin, Klein, and Oveis Gharan [2] showed that the partition-property approach cannot give a constant guarantee for general binary matroids. These barriers concern the specified algorithmic approaches rather than the existence of constant-competitive algorithms.

Constant guarantees were also known under other online models. Soto [50] studied random assignment of weights to elements, and Oveis Gharan and Vondrák [44] extended this line to adversarial arrival orders. Santiago, Sergeev, and Zenklusen [47] obtained a constant guarantee for random assignment and random arrival order without knowing the matroid. Jaillet, Soto, and Zenklusen [34] considered the free-order model, where the algorithm chooses the order in which elements are inspected.

Combinatorial allocations and group arrivals. Several works study secretary problems beyond single-element arrivals with additive weights. When individual elements arrive online and the goal is to maximize a global submodular function over the selected independent set, Feldman and Zenklusen [30] reduced the problem to the linear-objective case (see also [8,42]). Combined with recent algorithms for general matroids, this yields constant guarantees for nonnegative submodular objectives [49]; see also Rubinstein and Singla [46] for combinatorial prophet and secretary models optimizing XOS and subadditive functions over a single matroid. For settings where arriving agents receive bundles of elements, Kesselheim, Radke, Tönnis, and Vöcking [35] obtained an optimal $1/e$-competitive algorithm for online combinatorial auctions with submodular and XOS [27, 40] valuations, corresponding to disjoint bundle allocations without matroid constraints. Related random-order and sample-based allocation models for packing LPs and XOS valuations were studied in [22, 26, 36]. For unit-demand agents, the setting of [35] specializes to one-sided online weighted

bipartite matching, which Marinkovic, Soto, and Verdugo [43] extended to general independence systems on the items, including hypergraph matchings and matroid intersections admitting directed certifiers. In a different direction, Korula and Pál [38] introduced secretary problems with group arrivals, in which an adversary partitions the elements into groups that arrive in uniformly random order and the algorithm may select multiple elements from each arriving group, and studied this model for graph and hypergraph matching.

2 The Dimension-Invariant Algorithm

In this section, we describe our algorithm for the linear matroid secretary problem in the online-representation model. Let n be the number of elements, and fix a sample size $1 \leq k < n$. The first k arrivals form the sample and are rejected. In the classical rank-1 secretary algorithm, if the best element arrives in position $i > k$, then it is selected precisely when the best of the first $i-1$ elements appears in the sample, which happens with probability $k/(i-1)$. We seek the same conditional probability in a matroid: if X is the set of the first i observed elements and $e \in \mathrm{OPT}(X)$ arrives last, we want to accept e with probability $k/(i-1)$. As in the rank-1 case, choosing k appropriately and averaging over the arrival position yields a $1/e$ guarantee. We obtain these acceptance probabilities by solving linear programs that simultaneously enforce the desired acceptance probabilities and control, for every subspace L, the expected dimension of $L \cap \langle \mathrm{ALG} \rangle$.

2.1 The Dimension Invariant

When no confusion may arise, we identify an observed element with its representing vector. We use the notation $\langle \cdot \rangle$ for *linear span*. If X is a set of observed elements, then $\langle X \rangle$ denotes the span of their representing vectors, and $\langle e \rangle$ denotes the subspace spanned by the vector representing e. We write $W \leq V$ when W is a subspace of V, and for subspaces W_1, W_2 we write $W_1 + W_2 \coloneqq \langle W_1 \cup W_2 \rangle$. For a set X and an element e, we use the shorthand $X - e \coloneqq X \setminus \{e\}$ and $X + e \coloneqq X \cup \{e\}$. When an element e arrives, let X be the set of elements observed up to and including e. We call e *improving* if $e \in \mathrm{OPT}(X)$.

After the sample, the *state* of the algorithm consists of an observed set X, an accepted set ALG, and the span $\langle \mathrm{ALG} \rangle$ of the accepted vectors. In addition to its state, our algorithm maintains some auxiliary information for every set $Y \subseteq X$. We store a distribution μ_Y supported on the subspaces of $\langle Y \rangle$, for which we show the following: conditioned on Y being the first $|Y|$ arrivals, $\mu_Y(W)$ is the probability that $\langle \mathrm{ALG} \rangle = W$ after the elements of Y have been processed. We also store a conditional acceptance probability $p_Y(e, W)$ for every $e \in Y$ and $W \in \mathrm{supp}(\mu_{Y-e})$. We use $p_Y(e, W)$ when the elements of Y arrive first, e is the last among them, and the accepted elements span W immediately before its arrival. Recursive dependencies necessitate that we maintain this information for all sets $Y \subseteq X$, not merely those occurring as prefixes of the realized arrival order: μ_Y is evidently a function of p_Y and μ_{Y-e} for all $e \in Y$, and moreover we compute p_Y as a function of all the μ_{Y-e} in order to satisfy certain properties.

For every $Y \subseteq X$ with $|Y| \leq k$, if the elements of Y were the first $|Y|$ arrivals, the algorithm would reject all of them as part of the sample. Therefore, we set $\mu_Y(\{0\}) = 1$, $\mu_Y(W) = 0$ for every $W \neq \{0\}$, and $p_Y(e, W) = 0$ for every $e \in Y$ and $W \in \mathrm{supp}(\mu_{Y-e})$. Now suppose that $|X| = i > k$. For every $e \in \mathrm{OPT}(X)$, we want to accept e with probability $k/(i-1)$, conditional on X being the set of the first i arrivals and e arriving last among X. We maintain a particular invariant which ensures this remains possible, which we call the *Dimension Invariant Inequality.* In particular, for

every set X with $|X| \geq k$ whose distribution μ_X has been computed, we require that

$$\mathbb{E}_{W \sim \mu_X} [\dim(W \cap L)] \leq \left(1 - \frac{k}{|X|}\right) \dim(L) \quad \text{for every subspace } L. \tag{1}$$

To see why we use the factor $1 - k/|X|$, let $|Y| = i > k$ and suppose that the last element e to arrive from Y is improving. Apply (1) to $Y - e$ and $L = \langle e \rangle$. Since e is not a loop, $\langle e \rangle$ has dimension one. Moreover, for a random subspace W distributed according to μ_{Y-e}, $\dim(W \cap \langle e \rangle)$ is 1 if $e \in W$ and 0 otherwise. Hence (1) implies that $\mathbb{P}[e \notin W] \geq k/(i-1)$. Thus, before e arrives, it can be added to ALG without violating independence with at least the desired probability. We now need to choose the acceptance probabilities so that every improving element is accepted with exactly this probability and the Dimension Invariant Inequality continues to hold. Notice that it suffices to guarantee the Dimension Invariant Inequality for $L \leq \langle X \rangle$. This is because every $W \in \operatorname{supp}(\mu_X)$ satisfies $W \leq \langle X \rangle$, and thus $W \cap L = W \cap (L \cap \langle X \rangle)$. Therefore, applying (1) to $L \cap \langle X \rangle$ gives the inequality for an arbitrary ambient subspace L.

Fix the current observed set X, and let $Y \subseteq X$ with $|Y| = i > k$. Suppose that μ_A has been computed for every $A \subsetneq Y$. Suppose also that acceptance probabilities $p_Y(e, W) \in [0, 1]$ have been chosen for every $e \in Y$ and $W \in \operatorname{supp}(\mu_{Y-e})$, and write p for their collection. We define μ_Y as the distribution of the span obtained by choosing $e \in Y$ uniformly, sampling $W \sim \mu_{Y-e}$, and accepting e with probability $p_Y(e, W)$. If we reject e, the span remains W. If we accept e, the span becomes $W + \langle e \rangle$. Thus, for every $U \leq \langle Y \rangle$,

$$\mu_Y(U) = \frac{1}{i} \sum_{e \in Y} \sum_{W \in \operatorname{supp}(\mu_{Y-e})} \mu_{Y-e}(W) \left[(1 - p_Y(e, W)) \mathbf{1}[U = W] + p_Y(e, W) \mathbf{1}[U = W + \langle e \rangle] \right]. \tag{2}$$

Since each μ_{Y-e} is a probability distribution and $p_Y(e, W) \in [0, 1]$, the values in (2) are nonnegative and sum to one over all subspaces $U \leq \langle Y \rangle$. Thus, μ_Y is a probability distribution on the subspaces of $\langle Y \rangle$.

For every $e \in Y$ and every subspace L, let

$$u_e(L) = \mathbb{E}_{W \sim \mu_{Y-e}} [\dim(W \cap L)]$$

and

$$\Delta_e(W, L) = \dim((W + \langle e \rangle) \cap L) - \dim(W \cap L).$$

Thus $u_e(L)$ is the expected dimension of $W \cap L$ under μ_{Y-e}, and $\Delta_e(W, L)$ is the change in this dimension when W is replaced by $W + \langle e \rangle$. To impose (1) on μ_Y, define

$$g_p(L) \coloneqq -(i - k) \dim(L) + \sum_{e \in Y} u_e(L) + \sum_{e \in Y} \mathbb{E}_{W \sim \mu_{Y-e}} [p_Y(e, W) \Delta_e(W, L)]. \tag{3}$$

For each L, the function $g_p(L)$ is affine in p. The following lemma shows that μ_Y satisfies (1) for a subspace L if and only if $g_p(L) \leq 0$.

Lemma 1. *Let Y be a set with $|Y| = i > k$. Suppose that, for every $e \in Y$, μ_{Y-e} is a probability distribution supported on the subspaces of $\langle Y - e \rangle$. Let*

$$p = (p_Y(e, W) : e \in Y,\ W \in \operatorname{supp}(\mu_{Y-e}))$$

satisfy $p_Y(e, W) \in [0, 1]$ for every $e \in Y$ and $W \in \operatorname{supp}(\mu_{Y-e})$. Define μ_Y by (2) and g_p by (3). Then, for every subspace L,

$$\mathbb{E}_{U \sim \mu_Y} [\dim(U \cap L)] = \frac{1}{i}((i - k) \dim(L) + g_p(L)).$$

Consequently, for every subspace L, the Dimension Invariant Inequality for L holds if and only if $g_p(L) \le 0$. In particular, if

$$g_p(L) \le 0 \qquad \forall\, L \le \langle Y \rangle\,,$$

then μ_Y satisfies the Dimension Invariant Inequality.

Proof. By (2) and the definitions of $u_e(L)$ and $\Delta_e(W, L)$,

$$\begin{aligned}\mathbb{E}_{U\sim\mu_Y}\left[\dim(U \cap L)\right] &= \frac{1}{i}\sum_{e\in Y}\left(u_e(L) + \mathbb{E}_{W\sim\mu_{Y-e}}\left[p_Y(e, W)\Delta_e(W, L)\right]\right)\\ &= \frac{1}{i}\left((i-k)\dim(L) + g_p(L)\right),\end{aligned}$$

where the second equality follows from (3). Since $|Y| = i$, the Dimension Invariant Inequality for a fixed subspace L is

$$\mathbb{E}_{U\sim\mu_Y}\left[\dim(U \cap L)\right] \le \frac{i-k}{i}\dim(L).$$

By the identity above, this holds if and only if $g_p(L) \le 0$.

Therefore, if $g_p(L) \le 0$ for every $L \le \langle Y \rangle$, the Dimension Invariant Inequality holds for every $L \le \langle Y \rangle$. For an arbitrary subspace L, every $U \in \mathrm{supp}(\mu_Y)$ satisfies $U \cap L = U \cap (L \cap \langle Y \rangle)$. Applying the inequality to $L \cap \langle Y \rangle$ and using $\dim(L \cap \langle Y \rangle) \le \dim(L)$ gives the result for L. □

We will choose the acceptance probabilities so that $p_Y(e, W) = 0$ whenever $e \notin \mathrm{OPT}(Y)$, that is, whenever e would not be improving if it arrived last among the elements of Y. Under this restriction, (3) becomes

$$g_p(L) = -(i-k)\dim(L) + \sum_{e\in Y} u_e(L) + \sum_{e\in\mathrm{OPT}(Y)} \mathbb{E}_{W\sim\mu_{Y-e}}\left[p_Y(e, W)\Delta_e(W, L)\right].$$

2.2 The Acceptance LP and the Algorithm

Fix the current observed set X, and let $Y \subseteq X$ with $|Y| = i > k$. Suppose that μ_A has already been computed for every $A \subsetneq Y$. We choose the acceptance probabilities $p_Y(e, W)$, for $e \in Y$ and $W \in \mathrm{supp}(\mu_{Y-e})$, by solving the following feasibility LP, whose variables are precisely these quantities $p_Y(e, W)$. We write p for their collection.

$$\begin{aligned}\mathrm{LP}(Y):\quad & 0 \le p_Y(e, W) \le 1 && \forall\, e \in Y,\ W \in \mathrm{supp}(\mu_{Y-e}),\\ & p_Y(e, W) = 0 && \forall\, e \in Y,\ W \in \mathrm{supp}(\mu_{Y-e}) \text{ with } e \notin \mathrm{OPT}(Y) \text{ or } e \in W,\\ & \mathbb{E}_{W\sim\mu_{Y-e}}\left[p_Y(e, W)\right] = \frac{k}{i-1} && \forall\, e \in \mathrm{OPT}(Y),\\ & g_p(L) \le 0 && \forall\, L \le \langle Y \rangle\,.\end{aligned}$$

The first constraint ensures that the variables are valid probabilities. The second restricts acceptance to elements of $\mathrm{OPT}(Y)$ and assigns zero acceptance probability whenever $e \in W$, so that accepting e preserves independence. For each $e \in \mathrm{OPT}(Y)$, the third constraint sets the expected value of $p_Y(e, W)$ over $W \sim \mu_{Y-e}$ equal to the target value $k/(i-1)$. Finally, by Lemma 1, the last constraint is equivalent to requiring that the distribution μ_Y defined by (2) satisfies the Dimension Invariant Inequality. Since the representation is over a finite field, $\langle Y \rangle$ has only finitely many subspaces, so $\mathrm{LP}(Y)$ is a finite linear program (for arbitrary fields, see Remark 1 at the end of Section 3).

The following lemma shows that LP(Y) is feasible whenever the distributions μ_{Y-e} satisfy the Dimension Invariant Inequality (1). In other words, the inequality can indeed be maintained as an invariant by using acceptance probabilities feasible for this LP. Its proof is deferred to Section 3.

Lemma 2. *Let Y be a set with $|Y| = i > k$. Suppose that, for every $e \in Y$, the distribution μ_{Y-e} satisfies the Dimension Invariant Inequality* (1)*. Then* LP(Y) *is feasible.*

We fix a priori a rule for choosing a feasible solution of LP(Y), requiring the chosen solution to be rational whenever a rational feasible solution exists, and denote the chosen solution by p_Y. We then compute μ_Y from p_Y and the distributions μ_{Y-e} using (2). This completes the recursive construction of (μ_Y, p_Y). This fixed choice ensures that (μ_Y, p_Y) depends only on Y and the data revealed for its elements, and not on the order in which the elements of Y arrived. We will show in Lemma 3 that, throughout the execution, the relevant linear programs have rational coefficients and admit rational feasible solutions. Consequently, the rule above always selects rational acceptance probabilities.

Throughout its execution, our algorithm maintains a table indexed by subsets Y of the observed elements. For each such Y, the table stores the pair (μ_Y, p_Y), where

$$\mu_Y = \big(\mu_Y(U) : U \le \langle Y \rangle\big)$$

is a probability distribution on the subspaces of $\langle Y \rangle$, and

$$p_Y = \big(p_Y(f, W) : f \in Y,\ W \in \operatorname{supp}(\mu_{Y-f})\big)$$

is the collection of acceptance probabilities associated with Y. Once a pair (μ_Y, p_Y) is computed, it is stored and reused in all subsequent calls to UpdateDistributions. Thus, when a new element e arrives, the subroutine only needs to compute the pairs corresponding to sets Y that contain e.

We can now describe our online algorithm, detailed in Algorithm 1. After each arrival, the subroutine UpdateDistributions (Algorithm 2) computes the new entries of this table. The online algorithm then uses $p_X(e, \langle \mathrm{ALG} \rangle)$ to decide whether to accept the arriving element e. Since these probabilities are rational, the corresponding acceptance decisions can be implemented exactly.

Algorithm 1. Linear Matroid Secretary

```
Input: Number of elements n and sample size k
X ← ∅; ALG ← ∅
foreach arriving element e do
    Observe e's vector and its relative weight rank among the observed elements
    X ← X + e
    UpdateDistributions(X, e)
    if |X| > k then
        Accept e with probability p_X(e, ⟨ALG⟩)
        if e is accepted then
            ALG ← ALG + e
return ALG
```

Lemma 3. *Consider Algorithm 1 together with the subroutine* UpdateDistributions*. For every fixed set X, the algorithm is well-defined up to the processing of X, and the following properties hold:*

Algorithm 2. UpdateDistributions(X, e)

Input: The current observed set X, the newly arrived element e, together with the revealed vectors and relative weight order of the elements of X

```
foreach Y ⊆ X with e ∈ Y, in increasing order of |Y| do
    if |Y| ≤ k then
        Set μ_Y to the point mass at {0} and p_Y = 0
    else
        Compute OPT(Y) and enumerate the subspaces of ⟨Y⟩
        Solve LP(Y) using (μ_{Y−f})_{f∈Y}, and let p_Y be the chosen solution
        Compute μ_Y from (2)
    Store (μ_Y, p_Y)
```

(i) Conditional on X being the set of the first $|X|$ arrivals, μ_X is the distribution of $\langle \mathrm{ALG} \rangle$ after the elements of X have been processed.

(ii) ALG *is always independent.*

(iii) If $|X| \geq k$, then μ_X satisfies the Dimension Invariant Inequality (1)*.*

(iv) If $|X| = i > k$ and $e \in \mathrm{OPT}(X)$, then, conditional on X being the set of the first i arrivals and e arriving last among them, the algorithm accepts e with probability $k/(i-1)$.

(v) All entries of μ_X and p_X are rational.

Proof. We prove simultaneously, by induction on $|X|$, that the algorithm is well defined and satisfies all five properties.

As our base case, we take $|X| \leq k$. All elements of X belong to the sample and are rejected. Hence $\langle \mathrm{ALG} \rangle = \{0\}$ and μ_X is the point mass at $\{0\}$. Moreover, $p_X = 0$, so all entries of μ_X and p_X are rational, proving (v). Thus, μ_X is the distribution of $\langle \mathrm{ALG} \rangle$, proving (i), and ALG is independent, proving (ii). When $|X| = k$, (1) holds with equality, proving (iii).

Now we proceed by induction for X with $|X| = i > k$. First we show the algorithm is well defined up to the processing of X, and moreover satisfies property (v). Let a be the last element of X to arrive. When UpdateDistributions(X, a) processes X, every μ_{X-f} with $f \in X$ has already been computed: if $f = a$, it was computed before the arrival of a, while if $f \neq a$, the set $X - f$ has size $i - 1$ and, since sets are processed in increasing order of cardinality, it is processed earlier in the same call to UpdateDistributions. For $f \neq a$, the set $X - f$ need not be the set of the first $i - 1$ arrivals. However, by the fixed choice rule, (μ_{X-f}, p_{X-f}) depends only on $X - f$ and the data revealed for its elements, so the induction hypothesis still applies. Hence, by (iii) and (v), each μ_{X-f} satisfies the Dimension Invariant Inequality and has rational entries. Lemma 2 therefore implies that $\mathrm{LP}(X)$ is feasible. Moreover, $\mathrm{LP}(X)$ has rational coefficients, so it admits a rational feasible solution. By our fixed choice rule, p_X has rational entries, and (2) then implies that μ_X has rational entries. Thus, p_X and μ_X are well-defined, and (v) holds.

For property (i), fix $e \in X$ and condition on e being the last element to arrive among the elements of X. The relative order of the elements in $X - e$ is then uniformly random. Thus, by induction, immediately before the arrival of e, $\langle \mathrm{ALG} \rangle$ has distribution μ_{X-e}. Now fix $W \in \mathrm{supp}(\mu_{X-e})$. Conditional on $\langle \mathrm{ALG} \rangle = W$ immediately before the arrival of e, the algorithm rejects e with probability $1 - p_X(e, W)$, in which case $\langle \mathrm{ALG} \rangle$ remains equal to W, and accepts e with probability $p_X(e, W)$, in which case $\langle \mathrm{ALG} \rangle$ becomes $W + \langle e \rangle$. Therefore, conditional on e being the last element

to arrive, the probability that $\langle \mathrm{ALG} \rangle = U$ after processing e is

$$\sum_{W \in \mathrm{supp}(\mu_{X-e})} \mu_{X-e}(W) \left[(1 - p_X(e, W)) \mathbf{1}[U = W\}] + p_X(e, W) \mathbf{1}[U = W + \langle e \rangle] \right].$$

Every element of X is equally likely to arrive last. Averaging the expression above over $e \in X$ gives exactly (2) with $Y = X$. Hence μ_X is the distribution of $\langle \mathrm{ALG} \rangle$ after processing the elements of X. This proves (i).

We next prove that ALG remains independent (property (ii)). Let e be the last element of X to arrive. By induction, ALG is independent before e arrives. If e is accepted, the second constraint of $\mathrm{LP}(X)$ implies that $e \notin \langle \mathrm{ALG} \rangle$. Therefore, $\mathrm{ALG} + e$ is independent. This proves (ii).

For the Dimension Invariant Inequality (property (iii)), the last constraint of $\mathrm{LP}(X)$ gives

$$g_{p_X}(L) \leq 0 \qquad \forall\, L \leq \langle X \rangle .$$

By Lemma 1, μ_X satisfies (1) for every $L \leq \langle X \rangle$. As observed in Section 2.1, this implies the inequality for every subspace L. This proves (iii).

Finally, we prove that elements in $\mathrm{OPT}(X)$ are accepted with the desired probability $k/(i-1)$ conditioned on arriving last (Property (iv)). Fix $e \in \mathrm{OPT}(X)$ and condition on e being the last element to arrive among the elements of X. The relative order of the elements in $X - e$ is uniformly random. By induction, immediately before the arrival of e, $\langle \mathrm{ALG} \rangle$ has distribution μ_{X-e}. When $\langle \mathrm{ALG} \rangle = W$, the algorithm accepts e with probability $p_X(e, W)$, using fresh randomness. Therefore,

$$\mathbb{P}\left[e \in \mathrm{ALG} \mid e \text{ arrives last among } X\right] = \mathbb{E}_{W \sim \mu_{X-e}}\left[p_X(e, W)\right] = \frac{k}{i-1}, \tag{4}$$

where the last equality follows from the third constraint of $\mathrm{LP}(X)$. This proves (iv). □

2.3 The $1/e$ Guarantee

If $n = 1$ or $n = 2$, we do not run Algorithm 1. Instead, we accept the first non-loop element that arrives. Every element of $\mathrm{OPT}(E)$ is then selected with probability at least $1/n \geq 1/e$.

Hence assume that $n \geq 3$. We run Algorithm 1 with sample size $k = \lfloor n/e \rfloor$. Fix an element $e \in \mathrm{OPT}(E)$. For every $X \subseteq E$ containing e, we have $e \in \mathrm{OPT}(X)$. Indeed, since $e \in \mathrm{OPT}(E)$, the elements preceding e in the greedy order, with ties broken by the fixed rule, do not span e. Restricting those elements to X cannot make them span e.

Fix $i > k$ and condition on e arriving in position i and on X being the set of the first i arrivals. Then e arrives last among the elements of X and $e \in \mathrm{OPT}(X)$. By Lemma 3(iv), the conditional probability that e is accepted is $k/(i-1)$. Since the value $k/(i-1)$ does not depend on X, we may drop the conditioning on X. Thus, conditional on e arriving in position i, it is accepted with probability $k/(i-1)$. Since the arrival position of e is uniform,

$$\mathbb{P}\left[e \in \mathrm{ALG}\right] = \frac{1}{n} \sum_{i=k+1}^{n} \frac{k}{i-1} = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j}. \tag{5}$$

Since $1/x$ is decreasing,

$$k \sum_{j=k}^{n-1} \frac{1}{j} = 1 + k \sum_{j=k+1}^{n-1} \frac{1}{j} \geq 1 + k \int_{k+1}^{n} \frac{1}{x}\, dx = 1 + k \ln \frac{n}{k+1}.$$

It remains to show that the last expression is at least n/e. Since $k = \lfloor n/e \rfloor$, we have $ek \leq n < e(k+1)$. Let $h(x) = 1 + k \ln(x/(k+1)) - x/e$. On $[ek, e(k+1)]$ we have $h'(x) = k/x - 1/e \leq 0$, while $h(e(k+1)) = 0$. Hence $h(n) \geq 0$, and thus $1 + k \ln(n/(k+1)) \geq n/e$. Combining this with (5) gives $\mathbb{P}\left[e \in \mathrm{ALG}\right] \geq 1/e$. This completes the proof of Theorem 1.

3 Feasibility via Uncrossing

We prove Lemma 2 in this section. Throughout, we fix a set of elements X with $|X| = i > k$, together with a family of distributions

$$\boldsymbol{\mu} = \left(\mu_{X-e}\right)_{e \in X},$$

where each μ_{X-e} satisfies the Dimension Invariant Inequality (1). Our goal is to prove that $\mathrm{LP}(X)$ is feasible, thereby establishing Lemma 2. Let $P = P(X, \boldsymbol{\mu})$ be the polytope obtained from $\mathrm{LP}(X)$ by dropping the constraints $g_p(L) \leq 0$. Since X and $\boldsymbol{\mu}$ remain fixed throughout this section, we simply write P.

We first show that P is non-empty. We then show that there exists a point $p \in P$ that also satisfies $g_p(L) \leq 0$ for every $L \leq \langle X \rangle$. For $e \in \mathrm{OPT}(X)$, we have $\dim(\langle e \rangle) = 1$. Applying (1) to μ_{X-e} with $L = \langle e \rangle$ gives

$$\mathbb{P}_{W \sim \mu_{X-e}}\left[e \notin W\right] \geq \frac{k}{i-1}. \tag{6}$$

For such an e, set $p(e, W) = 0$ when $e \in W$, and otherwise set

$$p(e, W) = \frac{k}{(i-1)\mathbb{P}_{W' \sim \mu_{X-e}}\left[e \notin W'\right]}.$$

The inequality above ensures that $p(e, W) \leq 1$, while by construction $\mathbb{E}_{W \sim \mu_{X-e}}\left[p(e, W)\right] = k/(i-1)$. For $e \notin \mathrm{OPT}(X)$, set $p(e, W) = 0$ for every W. Thus $p \in P$, so P is non-empty.

It remains to find a point $p \in P$ such that $g_p(L) \leq 0$ for every $L \leq \langle X \rangle$. We begin with a structural property of g_p. Fix $p \in P$ and consider the following one-step experiment. Choose $e \in X$ uniformly, sample W according to μ_{X-e}, and accept e with probability $p(e, W)$. Let Z be the resulting spanned subspace. By Lemma 1, applied with $Y = X$, we have

$$g_p(L) = -(i-k)\dim(L) + i\,\mathbb{E}_p\left[\dim(Z \cap L)\right].$$

We claim that g_p is supermodular on the lattice of subspaces. Namely, for any $L, H \leq \langle X \rangle$,

$$g_p(L) + g_p(H) \leq g_p(L \cap H) + g_p(L + H).$$

Indeed, for every fixed subspace Z,

$$\dim(Z \cap L) + \dim(Z \cap H) = \dim((Z \cap L) + (Z \cap H)) + \dim(Z \cap L \cap H).$$

Together with

$$(Z \cap L) + (Z \cap H) \leq Z \cap (L + H),$$

this shows that $L \mapsto \dim(Z \cap L)$ is supermodular. Taking expectations preserves supermodularity, while $\dim(L)$ is modular. Therefore, g_p is supermodular.

We now use this supermodularity to reduce the problem to chains of subspaces.

Lemma 4. *Suppose that for every chain*

$$\mathcal{C} = \{L_1, \ldots, L_s\}, \qquad L_1 < \cdots < L_s \leq \langle X \rangle,$$

there exists a $p \in P$ such that $g_p(L_j) \leq 0$ for every $j \in \{1, \ldots, s\}$. Then there exists a single $p \in P$ such that $g_p(L) \leq 0$ for every $L \leq \langle X \rangle$.

Proof. Since the field is finite, $\langle X \rangle$ has finitely many subspaces. Let Λ be the probability simplex on these subspaces. By the von Neumann minimax theorem,

$$v \coloneqq \min_{p \in P} \max_{L \leq \langle X \rangle} g_p(L) = \max_{\lambda \in \Lambda} \min_{p \in P} \sum_L \lambda_L g_p(L). \tag{7}$$

Thus, the existence of a $p \in P$ such that $g_p(L) \leq 0$ for every $L \leq \langle X \rangle$ is equivalent to $v \leq 0$.

Since P is a polytope and $g_p(L)$ is affine in p, the minimum over P is attained at a vertex. Thus the map $\lambda \mapsto \min_{p \in P} \sum_L \lambda_L g_p(L)$ is the minimum of finitely many linear functions and is continuous. Its maximizers over the compact simplex Λ form a nonempty compact set. Choose λ among these maximizers to maximize $\sum_L \lambda_L (\dim(L))^2$. Suppose that two subspaces L, H, neither containing the other, have positive mass in λ, and let $t = \min\{\lambda_L, \lambda_H\}$. Move mass t from each of L and H to $L \cap H$ and $L + H$ and let λ' denote the resulting point in Λ. By the supermodularity of g_p, we have $g_p(L) + g_p(H) \leq g_p(L \cap H) + g_p(L + H)$ for every $p \in P$. Hence the weighted sum does not decrease for any p, and neither does its minimum over P. Since the original λ is optimal, the new point λ' is also optimal.

On the other hand, $\sum_L \lambda_L (\dim(L))^2$ increases by $2t(\dim(L) - \dim(L \cap H))(\dim(H) - \dim(L \cap H)) > 0$, a contradiction. Therefore, the chosen optimal λ is supported on a chain $\mathcal{C}$. By the hypothesis of the lemma, there exists a $p \in P$ with $g_p(L) \leq 0$ for every $L \in \mathcal{C}$. Thus, $v \leq \sum_L \lambda_L g_p(L) \leq 0$. □

By Lemma 4, it is enough to fix an arbitrary chain $L_1 < \cdots < L_s \leq \langle X \rangle$ and construct a point $p \in P$ such that $g_p(L_j) \leq 0$ for every $j \in \{1, \ldots, s\}$. We will embark on precisely that for the remainder of this section.

For an element $e \in \mathrm{OPT}(X)$, we define

$$\mathcal{F}_e = \{W \in \mathrm{supp}(\mu_{X-e}) \mid e \notin W\}.$$

These are precisely the states in which e can be accepted. We will need the following property.

Lemma 5. *Let e be an element in $\mathrm{OPT}(X)$. For every $W \in \mathcal{F}_e$ and every subspace L, we have*

$$\Delta_e(W, L) = \mathbf{1}[e \in W + L].$$

Proof. Recall that $\Delta_e(W, L) = \dim((W + \langle e \rangle) \cap L) - \dim(W \cap L)$. Use $\dim(A \cap B) = \dim(A) + \dim(B) - \dim(A + B)$, first with $(A, B) = (W + \langle e \rangle, L)$ and then with $(A, B) = (W, L)$:

$$\begin{aligned}
\dim((W + \langle e \rangle) \cap L) - \dim(W \cap L) &= (\dim(W + \langle e \rangle) + \dim(L) - \dim(W + \langle e \rangle + L)) \\
&\quad - (\dim(W) + \dim(L) - \dim(W + L)) \\
&= 1 - (\dim(W + \langle e \rangle + L) - \dim(W + L)) \\
&= 1 - \mathbf{1}[e \notin W + L] \\
&= \mathbf{1}[e \in W + L],
\end{aligned}$$

where the second equality follows from the fact that, since $W \in \mathcal{F}_e$, we have $e \notin W$ and thus $\dim(W + \langle e \rangle) - \dim(W) = 1$. □

Now we fix $e \in \mathrm{OPT}(X)$, and design the acceptance probabilities $p(e, .)$. For each j, let

$$\mathcal{H}_j = \{W \in \mathcal{F}_e \mid e \notin W + L_j\}.$$

By Lemma 5, accepting e does not increase $\dim(W \cap L_j)$ when $W \in \mathcal{H}_j$, and increases it by one when $W \in \mathcal{F}_e \setminus \mathcal{H}_j$. Since $L_1 < \cdots < L_s$, we have $\mathcal{H}_s \subseteq \cdots \subseteq \mathcal{H}_1$. Intuitively, $\mathcal{H}_s$ consists of the

"safest" states W for accepting e, in that doing so does not increase $\dim(W \cap L_j)$ for any subspace L_j in our chain. More generally, for $1 \le j < s$, the states in $\mathcal{H}_j \setminus \mathcal{H}_{j+1}$ are safe for $L_1, \ldots, L_j$ but unsafe for $L_{j+1}, \ldots, L_s$. Enumerate $\mathcal{F}_e$ as $W_1, \ldots, W_t$, in order of most to least safe, by listing first the states in $\mathcal{H}_s$, then those in $\mathcal{H}_{s-1} \setminus \mathcal{H}_s$, and so on, ending with $\mathcal{F}_e \setminus \mathcal{H}_1$ which are unsafe for all L_j. The order within each block is arbitrary. For every j, the states in $\mathcal{H}_j$ therefore form an initial segment of this enumeration. For $W \in \operatorname{supp}(\mu_{X-e})$ with $e \in W$, set $p(e, W) = 0$. For $\ell = 1, \ldots, t$, assign acceptance probabilities in that order:

$$p(e, W_\ell) = \min \left\{ 1, \frac{\left[\frac{k}{i-1} - \sum_{h<\ell} \mu_{X-e}(W_h)\right]^+}{\mu_{X-e}(W_\ell)} \right\}, \tag{8}$$

where $[x]^+ \coloneqq \max\{x, 0\}$. Since each W_ℓ has positive probability under μ_{X-e}, the denominator of the expression above is positive.

Recall that $e \in \mathrm{OPT}(X)$ is fixed throughout the above construction of the probabilities $p(e, .)$. By (6) and the definition of $\mathcal{F}_e$,

$$\mu_{X-e}(\mathcal{F}_e) = \mathbb{P}_{W \sim \mu_{X-e}}[e \notin W] \ge \frac{k}{i-1}.$$

Hence the greedy assignment in (8) assigns total acceptance probability exactly $k/(i-1)$ to e. We perform the construction above separately for each $e \in \mathrm{OPT}(X)$, and set $p(e, W) = 0$ for every $e \notin \mathrm{OPT}(X)$. This defines all the coordinates of a point p. Since, for each $e \in \mathrm{OPT}(X)$, the construction assigns total acceptance probability $k/(i-1)$ and sets $p(e, W) = 0$ whenever $e \in W$, we have $p \in P$.

Now fix $e \in \mathrm{OPT}(X)$. Since every $\mathcal{H}_j$ is an initial segment,

$$\sum_{W \in \mathcal{H}_j} \mu_{X-e}(W) p(e, W) = \min\left\{ \frac{k}{i-1}, \mu_{X-e}(\mathcal{H}_j) \right\}. \tag{9}$$

For every L in the chain, let $q_e^p(L) = \mathbb{E}_{W \sim \mu_{X-e}}[p(e, W)\Delta_e(W, L)]$ be the expected increase in $\dim(W \cap L)$ when we accept e according to p. Recall that $u_e(L) = \mathbb{E}_{W \sim \mu_{X-e}}[\dim(W \cap L)]$, so $u_e(L) + q_e^p(L)$ is the expected dimension of the intersection of L with the span of the accepted elements after processing e. We will establish an upper bound on $u_e(L) + q_e^p(L)$, before summing over all e to show that $g_p(L) \le 0$. First, we re-express $q_e^p(L)$ as follows.

$$\begin{aligned}
q_e^p(L) &= \sum_{W \in \mathcal{F}_e} \mu_{X-e}(W) p(e, W) \Delta_e(W, L) \\
&= \sum_{W \in \mathcal{F}_e : e \in W + L} \mu_{X-e}(W) p(e, W) \\
&= \sum_{W \in \mathcal{F}_e} \mu_{X-e}(W) p(e, W) - \sum_{W \in \mathcal{F}_e : e \notin W + L} \mu_{X-e}(W) p(e, W) \\
&= \frac{k}{i-1} - \min\left\{ \frac{k}{i-1}, \mu_{X-e}(\{W \in \mathcal{F}_e \mid e \notin W + L\}) \right\} \\
&= \left(\frac{k}{i-1} - \mu_{X-e}(\{W \in \mathcal{F}_e \mid e \notin W + L\}) \right)^+ \\
&= \left(\frac{k}{i-1} - \mathbb{P}_{W \sim \mu_{X-e}}[e \notin W + L] \right)^+,
\end{aligned} \tag{10}$$

where the second equality follows from Lemma 5, and the fourth follows from (9).

Suppose first that $e \notin L$. Similarly to Lemma 5, the modularity of dimension on subspaces easily yields

$$\dim(W \cap (L + \langle e \rangle)) - \dim(W \cap L) = \mathbf{1}[e \in W + L].$$

Taking expectations and rearranging, we obtain

$$u_e(L) + \frac{k}{i-1} - \mathbb{P}_{W \sim \mu_{X-e}}[e \notin W + L] = u_e(L + \langle e \rangle) - \left(1 - \frac{k}{i-1}\right). \tag{11}$$

Combining (10) and (11) gives

$$\begin{aligned} u_e(L) + q_e^p(L) &= u_e(L) + \left[\frac{k}{i-1} - \mathbb{P}_{W \sim \mu_{X-e}}[e \notin W + L]\right]^+ \\ &= \max\left\{u_e(L), u_e(L + \langle e \rangle) - \left(1 - \frac{k}{i-1}\right)\right\}. \end{aligned} \tag{12}$$

The Dimension Invariant Inequality (1) for $X - e$, when applied to L and $L + \langle e \rangle$, gives

$$\begin{aligned} u_e(L) &\le \left(1 - \frac{k}{i-1}\right)\dim(L), \\ u_e(L + \langle e \rangle) &\le \left(1 - \frac{k}{i-1}\right)\dim(L + \langle e \rangle) = \left(1 - \frac{k}{i-1}\right)(\dim(L) + 1). \end{aligned}$$

Combining this with (12) yields

$$u_e(L) + q_e^p(L) = \max\left\{u_e(L), u_e(L + \langle e \rangle) - \left(1 - \frac{k}{i-1}\right)\right\} \le \left(1 - \frac{k}{i-1}\right)\dim(L). \tag{13}$$

If $e \in L$, Lemma 5 gives $\Delta_e(W, L) = 1$ for every $W \in \mathcal{F}_e$. Thus $q_e^p(L) = k/(i-1)$, and (1) gives $u_e(L) \le (1 - \frac{k}{i-1})\dim(L)$. Together with (13) for $e \notin L$, this gives

$$u_e(L) + q_e^p(L) \le \left(1 - \frac{k}{i-1}\right)\dim(L) + \frac{k}{i-1} \cdot \mathbf{1}[e \in L].$$

For $e \notin \mathrm{OPT}(X)$, we have $p(e, W) = 0$ for every W, and the Dimension Invariant Inequality gives

$$u_e(L) \le \left(1 - \frac{k}{i-1}\right)\dim(L).$$

Since $\mathrm{OPT}(X)$ is independent, $|\mathrm{OPT}(X) \cap L| \le \dim(L)$. Summing these bounds over all elements in X gives

$$\begin{aligned} \sum_{e \in X} u_e(L) + \sum_{e' \in \mathrm{OPT}(X)} q_{e'}^p(L) &\le i\left(1 - \frac{k}{i-1}\right)\dim(L) + \frac{k}{i-1} \cdot |\mathrm{OPT}(X) \cap L| \\ &\le \left(i\left(1 - \frac{k}{i-1}\right) + \frac{k}{i-1}\right)\dim(L) = (i-k)\dim(L). \end{aligned}$$

Therefore $g_p(L) \le 0$ for every L in the chain. Lemma 4 gives a single $p \in P$ satisfying all the constraints of $\mathrm{LP}(X)$. This proves Lemma 2.

Remark 1. Ignoring computational issues, Theorem 1 also holds in the online-representation model over arbitrary fields, including infinite fields. Although the observed span is finite-dimensional, it may then have infinitely many subspaces. We show that the LP still has finitely many variables and distinct constraints, and that the proof of Lemma 4 continues to apply.

Fix an observed set Y with $|Y| > k$, and let $\mathcal{S}_Y = \{\langle Z\rangle : Z \subseteq Y\}$, which is finite. By induction, $\mathrm{supp}(\mu_{Y-e}) \subseteq \mathcal{S}_Y$ for every $e \in Y$. Thus there are finitely many variables $p_Y(e, W)$. Moreover, $W + \langle e\rangle \in \mathcal{S}_Y$ for every such pair. The constraint indexed by $L \leq \langle Y\rangle$ therefore depends only on $\dim(L)$ and its intersection dimensions with the subspaces in $\mathcal{S}_Y$. We record this information in the signature

$$\sigma_Y(L) = \big(\dim(L),\, (\dim(W \cap L))_{W \in \mathcal{S}_Y}\big).$$

Each entry is an integer between 0 and $|Y|$, so only finitely many signatures occur. Subspaces with the same signature define the same constraint. We can therefore retain one representative subspace for each signature, obtaining a finite LP.

The proof of Lemma 4 now applies to these representatives. In its uncrossing step, suppose that two chosen representatives L, H have positive λ-mass and neither contains the other. We move equal amounts of mass from L, H to the chosen representatives with signatures $\sigma_Y(L \cap H)$ and $\sigma_Y(L{+}H)$, respectively. Replacing a subspace with another having the same signature preserves both its LP constraint and its dimension. As in the proof of Lemma 4, the minimax objective does not decrease, while $\sum_L \lambda_L(\dim(L))^2$ strictly increases. The construction in (8) also works without change. The coefficients of the resulting finite LP are still rational, so Lemma 3(v) continues to hold, and in Algorithm 2 we enumerate one representative subspace per signature instead of all subspaces of $\langle Y\rangle$.

4 Finitary Modular Extensions

We now turn to the known-matroid model. For linear matroids, the result follows from Theorem 1, since every finite linear matroid admits a representation over a finite field. We then show that the same guarantee holds more generally for matroids admitting a finitary modular extension.

Corollary 1. *In the known-matroid model, every linear matroid admits an ordinal* $1/e$*-probability-competitive algorithm.*

Proof. By a theorem of Rado [45], every finite matroid representable over some field is representable over a finite field. Since the matroid is known and we allow unbounded computational power, we may enumerate finite fields and matrices with columns indexed by E, checking each candidate against the independence structure of the matroid, until we find such a representation. We do not require this search to be efficient. Once a finite-field representation is fixed, Theorem 1 applies. □

Thus the representation need not be supplied as part of the input in the known-matroid model. In particular, the corollary applies to every finite linear matroid, regardless of the field over which it is initially known to be representable.

For a matroid N, we write cl_N for its *closure operator*. A set F is a *flat* if $\mathrm{cl}_N(F) = F$. For two flats F_1, F_2, their *meet* is $F_1 \cap F_2$ and their *join* is $F_1 \vee F_2 \coloneqq \mathrm{cl}_N(F_1 \cup F_2)$. The pair F_1, F_2 is *modular* if $r_N(F_1) + r_N(F_2) = r_N(F_1 \cap F_2) + r_N(F_1 \vee F_2)$. The matroid is *modular* if every pair of its flats is modular. A possibly infinite matroid is *finitary* if every circuit is finite. We say that a finite matroid M on E admits a *finitary modular extension* if there is a finitary modular matroid M' on a ground set containing E such that $M'|E = M$. We only use the restriction of M' to

$\mathrm{cl}_{M'}(E)$, which has finite rank and is again modular. Every circuit of this restriction has at most $r_{M'}(E)+1$ elements, so the restriction is finitary.

The proof of Theorem 1 in fact uses less than a vector-space representation. The arguments only require the modular rank identities for flats. This yields the following more general consequence.

Corollary 2. *Let M be a known matroid admitting a finitary modular extension. Then M admits an ordinal $1/e$-probability-competitive algorithm.*

Proof. Suppose first that an extension M' is fixed and available to the algorithm. We work with the flats of M' contained in $\mathrm{cl}_{M'}(E)$, ordered by inclusion. Since E is finite, each such flat has rank at most $r_{M'}(E) < \infty$.

The construction of Section 2 carries over by replacing vector spans with closures in M', sums of subspaces with joins of flats, and dimension with $r_{M'}$. Thus, the state corresponding to an accepted set A is $\mathrm{cl}_{M'}(A)$, the initial state is $\mathrm{cl}_{M'}(\emptyset)$, and the Dimension Invariant Inequality becomes

$$\mathbb{E}_{W\sim\mu_X}\left[r_{M'}(W\cap L)\right] \le \left(1-\frac{k}{|X|}\right) r_{M'}(L)$$

for every flat $L \subseteq \mathrm{cl}_{M'}(E)$.

For every fixed flat W, the function $F \mapsto r_{M'}(W\cap F)$ is supermodular. Indeed, $(W\cap F_1)\vee(W\cap F_2)\subseteq W\cap(F_1\vee F_2)$, and we can apply modularity to the pairs $W\cap F_1, W\cap F_2$ and F_1, F_2. We can therefore repeat the uncrossing argument from Section 3.

For the construction along a fixed chain, let L be a flat in the chain. Modularity gives

$$\begin{aligned} r_{M'}(\mathrm{cl}_{M'}(W\cup\{e\})\cap L) &= r_{M'}(W\cap L)+\mathbf{1}[e\in\mathrm{cl}_{M'}(W\cup L)] && \text{if } e\notin W,\\ r_{M'}(W\cap \mathrm{cl}_{M'}(L\cup\{e\})) &= r_{M'}(W\cap L)+\mathbf{1}[e\in\mathrm{cl}_{M'}(W\cup L)] && \text{if } e\notin L. \end{aligned}$$

Also, $\mathrm{OPT}(X)$ is independent in $M = M'|E$, so $|\mathrm{OPT}(X)\cap L| \le r_{M'}(L)$. We can now repeat the construction along a fixed chain and the proof of (13), using closures and rank in place of spans and dimension.

There may be infinitely many flats contained in $\mathrm{cl}_{M'}(E)$. As in Remark 1, we use signatures to show that the acceptance LP nevertheless has only finitely many distinct constraints. For a fixed observed set X, every state that can occur is of the form $\mathrm{cl}_{M'}(Z)$ for some $Z\subseteq X$. Hence the set

$$\mathcal{S}_X = \{\mathrm{cl}_{M'}(Z)\colon Z\subseteq X\}$$

is finite. For a flat $L\subseteq\mathrm{cl}_{M'}(E)$, define its signature relative to X by

$$\sigma_X(L) = \Big(r_{M'}(L), (r_{M'}(W\cap L))_{W\in\mathcal{S}_X}\Big).$$

Every quantity appearing in the constraint $g_p(L)\le 0$ is determined by this signature. Indeed, both a predecessor state W and the state obtained after accepting an element belong to $\mathcal{S}_X$. Since all ranks in $\sigma_X(L)$ are integers between 0 and $r_{M'}(E)$, only finitely many signatures can occur. Thus there are only finitely many distinct constraints.

The chain-reduction argument can now be applied to these finitely many constraint types. Choose one representative flat for each signature. If two representatives carrying positive mass in an optimal minimax solution are incomparable, move their mass, exactly as in the proof of Lemma 4, to the chosen representatives of the signatures of their intersection and join. Flats with the same signature define the same constraint and have the same rank, so supermodularity ensures that the minimax objective does not decrease, while modularity gives the same strict increase in

the secondary objective. Therefore an optimal solution can again be supported on a chain. The construction for a fixed chain uses only the Dimension Invariant Inequality and the properties established above, so the remainder of the proof of Lemma 2 carries over unchanged. We therefore obtain an ordinal randomized policy with the same $1/e$ probability guarantee as in Theorem 1.

This policy was constructed using the extension M'. Finally, we show that the extension need not be given to the algorithm. Since M is finite, there are only finitely many deterministic ordinal online policies that always maintain independence in M. A history is determined by the ordered sequence of labels seen so far, their relative weight order, and the previous accept/reject decisions. Fixing all random choices of the policy constructed above in advance therefore induces a probability distribution over this finite set of deterministic policies.

We can recover a suitable distribution using only the known matroid M. Consider the finite LP that chooses a probability distribution over deterministic ordinal policies and maximizes a variable t. For every total order on E and every element of the corresponding optimal basis, require that its selection probability, averaged over all arrival orders, be at least t. All coefficients of this LP are determined by M. The randomized policy constructed from M' provides a feasible distribution with $t \geq 1/e$, so the optimum of the LP is at least $1/e$. We may therefore solve this LP before the arrivals, sample a deterministic policy from the resulting distribution, and follow it online. The resulting algorithm depends only on the known matroid M and has the required probability guarantee. □

For finite matroids, admitting a finitary modular extension is equivalent to being fully modular extendable. The existence of such an extension for every fully modular extendable matroid follows from [9, Lemma 3.2]. Hence Corollary 2 applies precisely to fully modular extendable matroids. By [9, Theorem 3.4], these are exactly the matroids whose connected components either have rank 3 or are skew-representable. In particular, Corollary 2 covers every matroid of rank at most 3, since matroids of rank at most 2 are linear. Its scope strictly extends beyond linear matroids: the rank-3 non-Pappus and non-Desargues matroids are not linear, but both satisfy its hypotheses. By contrast, the Vámos matroid is connected, has rank 4, and is not skew-representable, so it does not admit a finitary modular extension.

5 XOS Valuations and Group Arrivals

A valuation $f : 2^E \to \mathbb{R}_{\geq 0}$ is *XOS* (*fractionally subadditive*) [27,40] if there exists a finite nonempty family $\mathcal{Q} \subseteq \mathbb{R}^E_{\geq 0}$ of nonnegative additive valuations such that, for all $D \subseteq E$,

$$f(D) = \max_{q \in \mathcal{Q}} \sum_{e \in D} q_e.$$

Each $q \in \mathcal{Q}$ is an additive *clause*. The following simple observation will be useful in our analysis. For any $B \subseteq E$, fix a *supporting* clause q with $f(B) = \sum_{e \in B} q_e$; then $f(D) \geq \sum_{e \in D} q_e$ for all $D \subseteq E$. Consequently, if $T \subseteq B$ is random and $\mathbb{P}[e \in T] = \alpha$ for all $e \in B$, then

$$\mathbb{E}[f(T)] \geq \mathbb{E}\left[\sum_{e \in T} q_e\right] = \alpha f(B). \tag{14}$$

Observe that additive valuations are XOS with $|\mathcal{Q}| = 1$. A unit-demand valuation is also XOS: for nonnegative weights $(w_e)_{e \in E}$, define $f(\emptyset) = 0$ and $f(D) = \max_{e \in D} w_e$ for $D \neq \emptyset$. For $E \neq \emptyset$, take $\mathcal{Q}$ to consist of the clauses that assign weight w_e to one element e and 0 to all other elements. If $E = \emptyset$, take $\mathcal{Q}$ to consist of the empty clause.

XOS Matroid Secretary. Let $M = (E, \mathcal{I})$ be a known matroid, and let N be a set of n agents, where n is known in advance. The agents arrive in uniformly random order. Each agent a reveals an eligibility set $E_a \subseteq E$ and an XOS valuation $f_a : 2^{E_a} \to \mathbb{R}_{\geq 0}$. Upon arrival, the algorithm irrevocably assigns a bundle $T_a \subseteq E_a$ to agent a. Eligibility sets may overlap, but the assigned bundles must be pairwise disjoint, and their union must be independent in M. The goal is to maximize $\sum_{a \in N} f_a(T_a)$. We remark that all eligibility sets and valuations are fixed before the arrival order is drawn, and we allow unbounded computation and value queries.

For a set of agents $X \subseteq N$, we represent an allocation by a set $S = \{(a, T_a) : a \in X\}$, where $T_a \subseteq E_a$. Write

$$\overline{S} = \bigcup_{a \in X} T_a \qquad \text{and} \qquad F(S) = \sum_{a \in X} f_a(T_a)$$

for its set of assigned elements and its value. Throughout this section, $\mathrm{OPT}(X)$ and $\mathrm{ALG}(X)$ denote allocations for the offline optimal solution and the algorithm, respectively. In particular, $\mathrm{ALG}(X) = \{(a, T_a) : a \in X\}$ after the agents of X have been processed, including pairs with empty bundles. Once agent a has arrived, the algorithm keeps E_a and can query f_a throughout the execution. For every set X of observed agents, let

$$\mathrm{OPT}(X) = \{(a, B_a(X)) : a \in X\}$$

be an offline optimal allocation for the subinstance on X, chosen by a fixed deterministic tie-breaking rule. We compute $\mathrm{OPT}(X)$ over all feasible allocations for the agents in X, using their eligibility sets and valuations. Its value is $F(\mathrm{OPT}(X)) = \sum_{a \in X} f_a(B_a(X))$.

When M is the free matroid on E, we only require the assigned bundles to be pairwise disjoint. We then obtain the online combinatorial auction problem, for which Kesselheim, Radke, Tönnis, and Vöcking [35] obtained an optimal $1/e$ guarantee for submodular valuations.[1] Our Theorem 2 generalizes their guarantee from the free matroid to every known linear matroid. The model also includes the following two settings: *group matroid secretary* and *agent matroid secretary*, in which each agent receives at most one element.

Group Matroid Secretary. An adversary partitions E into n groups $(G_a)_{a \in N}$. The groups arrive in uniformly random order; n is known in advance, but the partition is not. When a group arrives, its elements and their nonnegative weights are revealed. The algorithm may accept any subset of the arriving group, as long as the union of all accepted elements remains independent in M. Korula and Pál [38] introduced secretary problems with groups and studied this model for matching.

This is a special case of the XOS model: take $E_a = G_a$ and $f_a(D) = \sum_{e \in D} w_e$ for every $D \subseteq G_a$. Order E by nonincreasing weight, using the fixed tie-breaking rule from Section 1. For every observed set of groups $X \subseteq N$, let $I(X)$ be the greedy maximum-weight basis in $\bigcup_{a \in X} G_a$ under this order, and set $B_a(X) = I(X) \cap G_a$. We choose $\mathrm{OPT}(X) = \{(a, B_a(X)) : a \in X\}$. Then $\overline{\mathrm{OPT}(X)} = I(X)$ and $F(\mathrm{OPT}(X)) = w(I(X))$.

Agent Matroid Secretary. Each agent a reveals its eligible elements E_a and nonnegative weights $(w_{a,e})_{e \in E_a}$. At most one element may be assigned to each agent, each element may be assigned at most once, and the set of assigned elements must be independent in M. Define $f_a(\emptyset) = 0$ and $f_a(D) = \max_{e \in D} w_{a,e}$ for every nonempty $D \subseteq E_a$. In any feasible allocation, we can replace

[1] Their proof uses Proposition 8 of [35] to bound the expected value of a random subset. The supporting-clause argument in (14) proves the same bound for XOS valuations, also when each inclusion probability is at least α. Their analysis therefore extends to XOS valuations.

each nonempty bundle by one of its maximum-weight elements without changing the total value or violating feasibility. We therefore choose each $\mathrm{OPT}(X)$ so that $|B_a(X)| \le 1$ for every $a \in X$.

A special case is weighted bipartite matching with one-sided vertex arrivals: offline vertices are known in advance, online vertices arrive in random order, and reveal their incident edges and weights. Each online vertex must be matched immediately to an available offline neighbor or rejected. This corresponds to taking M to be the free matroid on the offline vertices. Kesselheim, Radke, Tönnis, and Vöcking [35] gave a $1/e$-competitive algorithm for this problem. Marinkovic, Soto, and Verdugo [43] extended this online assignment model to general independence systems on the items, including hypergraph matchings and matroid intersections admitting directed certifiers.

Approximation guarantees. The following theorem is the main result of this section.

Theorem 2. *For every known linear matroid, the XOS Matroid Secretary Problem admits an online algorithm that always produces a feasible allocation and satisfies*

$$\mathbb{E}[F(\mathrm{ALG})] \ge \frac{1}{e} F(\mathrm{OPT}(N)).$$

The same guarantee holds without prior knowledge of the matroid when a representation over a known finite field is revealed with the agents, in a common coordinate system.

The algorithm used to prove Theorem 2 also gives the following corollaries for group and agent arrivals.

Corollary 3. *For Group Matroid Secretary on a known linear matroid, there is an online algorithm that accepts each element of a fixed greedy optimum with probability at least $1/e$. The same guarantee holds without prior knowledge of the matroid when a representation over a known finite field is revealed with the groups, in a common coordinate system.*

Corollary 4. *For Agent Matroid Secretary on a known linear matroid, there is an online algorithm whose expected value is at least $1/e$ times the optimum.*

To prove Theorem 2, we extend Algorithm 1 to agent arrivals. In that algorithm, we decide whether to accept or reject the arriving element. Here, when agent a arrives, we assign it a subset of its *optimal* bundle $B_a(X)$ in $\mathrm{OPT}(X)$, where X is the set of agents observed so far. In Section 5.1, we establish the Dimension Invariant Inequality we need in the XOS matroid secretary context. Then, in Section 5.2, we show how to construct the linear program that encodes the distribution we use to sample at each iteration of the selection algorithm, and in Section 5.3 we show its feasibility via an uncrossing argument. Finally, in Section 5.4, we put things together to conclude Theorem 2, and then we prove the approximation guarantees in Corollary 3 and Corollary 4.

For Sections 5.1–5.3, fix a representation of M in a finite-dimensional vector space V over a finite field. We use the same notation for an element and its representing vector, but retain the element's label. Distinct elements may have the same representing vector. In the proof of Theorem 2, we compute such a representation from the known matroid before the arrivals, as in Corollary 1.

5.1 The Dimension Invariant for Bundles

Fix a sample size $k \in \{1, \dots, n-1\}$. For every set of observed agents X, let $V_X \coloneqq \langle \bigcup_{a \in X} E_a \rangle$. As in Section 2.1, we recursively construct a distribution μ_X supported on the subspaces of V_X. Conditional on X being the set of the first $|X|$ arrivals, μ_X will be the distribution of $\left\langle \overline{\mathrm{ALG}} \right\rangle$ after processing those agents.

Agents in the sample receive empty bundles, so for $|X| \le k$, we set $\mu_X = \delta_{\{0\}}$, the point mass at the zero subspace. For $|X| \ge k$, we require the following *Dimension Invariant Inequality*:

$$\mathbb{E}_{W\sim\mu_X}[\dim(W \cap L)] \le \left(1 - \frac{k}{|X|}\right)\dim(L), \quad \text{for every } L \le V. \tag{15}$$

This is (1) with the number of observed agents in place of the number of observed elements. As in Section 2.1, it suffices to impose (15) for $L \le V_X$. Indeed, every $W \in \operatorname{supp}(\mu_X)$ satisfies $W \cap L = W \cap (L \cap V_X)$ and $\dim(L \cap V_X) \le \dim(L)$.

Consider the arrival of agent a in position $i > k$, and let X be the set of the first i agents. Conditional on X and a, the agents in $X - a$ arrive in uniformly random order. Let $W = \left\langle \overline{\mathrm{ALG}} \right\rangle$ be the span of the elements assigned before a arrives, so $W \sim \mu_{X-a}$.

Put $\alpha = k/(i-1)$. We seek a random subset $R \subseteq B_a(X)$ that includes each element with probability α; recall that $B_a(X) \subseteq E_a$ is such that $(a, B_a(X)) \in \mathrm{OPT}(X)$. Even if each element can be added on its own, we may be unable to add them together. For example, if u, v are linearly independent and $W = \langle u + v\rangle$, adding either u or v increases the dimension by one. Adding both still increases it by only one. To preserve feasibility, we also require that $\overline{\mathrm{ALG}} \cup R$ be independent and that R contain no previously assigned element. Since the previously assigned elements are independent and span W, these requirements are equivalent to

$$\dim(W + \langle R\rangle) = \dim(W) + |R|. \tag{16}$$

For each $e \in B_a(X)$, the probability $\mathbb{P}[e \in R]$ is taken over $W \sim \mu_{X-a}$ and the randomized choice of R given W. Since f_a is XOS, we can apply (14) to the bundle $B_a(X)$ and its random subset R. Each element belongs to R with probability α, so

$$\mathbb{E}[f_a(R)] \ge \alpha f_a(B_a(X)). \tag{17}$$

We will use (17) to bound the expected value assigned at each arrival in the proof of Theorem 2.

Since $B_a(X)$ is independent, every $D \subseteq B_a(X)$ satisfies $\dim(\langle D\rangle) = |D|$. Applying (15) to μ_{X-a} with $L = \langle D\rangle$ therefore gives

$$\mathbb{E}[\dim(W \cap \langle D\rangle)] \le (1-\alpha)|D| \qquad \text{for every } D \subseteq B_a(X), \tag{18}$$

where we used $|X - a| = i - 1$ and $\alpha = k/(i-1)$. The following lemma uses the bounds in (18) to construct R satisfying (16) and $\mathbb{P}[e \in R] = \alpha$ for every $e \in B_a(X)$.

Lemma 6. *Let $\alpha \in [0,1]$, and let W be a random subspace of V such that, for every $L \le V$,*

$$\mathbb{E}[\dim(W \cap L)] \le (1-\alpha)\dim(L). \tag{19}$$

For every independent set C of vectors, there is a randomized rule that, after observing W, selects $R \subseteq C$ satisfying $\dim(W + \langle R\rangle) = \dim(W) + |R|$ and $\mathbb{P}[c \in R] = \alpha$ for every $c \in C$.

Proof. Fix an independent set C, and let ν be the distribution of W. For each W, let M_W be the matroid on C with rank function given, for $D \subseteq C$, by

$$r_W(D) = \dim(W + \langle D\rangle) - \dim(W) = \dim(\langle D\rangle) - \dim(W \cap \langle D\rangle). \tag{20}$$

To see that r_W is a matroid rank function, introduce a set B_W of new elements, disjoint from C, whose representing vectors form a basis of W. Let r be the rank function of the linear matroid on

$C \cup B_W$. Contracting B_W leaves ground set C and gives rank $r(D \cup B_W) - r(B_W) = r_W(D)$ for every $D \subseteq C$. A set $R \subseteq C$ is independent in M_W exactly when it satisfies (16).

Let $P(M_W)$ be the matroid independence polytope, that is, the convex hull of the incidence vectors of the independent sets of M_W. This is an integral polymatroid with rank function r_W. Define

$$P = \left\{ \sum_W \nu(W) z^W : \text{ for each } W, \text{ the vector } z^W \text{ belongs to } P(M_W) \right\}.$$

For each $z^W \in P(M_W)$, we can select a random independent set of M_W with inclusion probabilities z^W. Thus P consists of the inclusion probabilities we can obtain after averaging over W. It suffices to show that $\alpha \mathbf{1}_C \in P$.

When we add polymatroids with nonnegative weights, we add their rank functions with the same weights; see, e.g., [48, Theorem 44.6]. Thus, for polymatroids P_i on C with rank functions r_i and weights $\lambda_i \geq 0$,

$$\sum_i \lambda_i P_i = \left\{ x \in \mathbb{R}^C_{\geq 0} : \sum_{c \in D} x_c \leq \sum_i \lambda_i r_i(D) \quad \forall D \subseteq C \right\}.$$

Applying this identity to $P(M_W)$ with weights $\nu(W)$ gives

$$P = \left\{ x \in \mathbb{R}^C_{\geq 0} : \sum_{c \in D} x_c \leq \mathbb{E}[r_W(D)] \quad \forall D \subseteq C \right\}. \tag{21}$$

Thus P is a polymatroid with rank function $\mathbb{E}[r_W(\cdot)]$. This function need not be integer-valued, so P need not be a matroid polytope. We now check the inequalities in (21) for $x = \alpha \mathbf{1}_C$. Fix $D \subseteq C$. Since C is independent, $\dim(\langle D \rangle) = |D|$. Apply (19) with $L = \langle D \rangle$. Using (20), we obtain

$$\mathbb{E}[r_W(D)] = |D| - \mathbb{E}[\dim(W \cap \langle D \rangle)] \geq |D| - (1 - \alpha)|D| = \alpha |D|.$$

Since $\sum_{c \in D} x_c = \alpha |D|$, the inequality in (21) holds for every $D \subseteq C$. Hence $\alpha \mathbf{1}_C \in P$.

For each W, choose $z^W \in P(M_W)$ such that $\sum_W \nu(W) z^W = \alpha \mathbf{1}_C$. For each W, express z^W as a convex combination of incidence vectors of independent sets of M_W. After observing W, choose one of these independent sets as R, with probabilities given by the coefficients. Then $\dim(W + \langle R \rangle) = \dim(W) + |R|$ and $\mathbb{P}[c \in R \mid W] = z^W_c$ for every $c \in C$. Hence

$$\mathbb{P}[c \in R] = \sum_W \nu(W) z^W_c = \alpha \qquad \text{for every } c \in C. \qquad \square$$

Lemma 6 extends the acceptance rule of Section 3 to bundles. For a single element, (6) gives $\mathbb{P}[e \notin W] \geq \alpha$, and that rule accepts e with probability α. The lemma gives the corresponding choice for $B_a(X)$: each element is included with probability α, and the selected elements satisfy (16) together. The LP in Section 5.2 will also impose (15) on the resulting distribution μ_X.

5.2 The Allocation LP and the Algorithm

We now construct μ_X recursively from the distributions for smaller sets of agents. As in Section 2.2, we use a linear program to choose the assignment probabilities at each step. For $|X| \leq k$, we use $\mu_X = \delta_{\{0\}}$ and store $p_X = 0$. For $|X| = i > k$, suppose that μ_Y has been computed for every proper subset Y of X, and recall that $\alpha = k/(i-1)$. For every $a \in X$, $W \in \mathrm{supp}(\mu_{X-a})$, and $T \subseteq B_a(X)$, we introduce a variable $p(a, W, T)$ representing the probability of assigning T to a when a arrives last among X and the previously assigned elements span W.

When $p(a, W, \cdot)$ is a probability distribution for each a and W, define the random subspace Z_p as follows: choose $a \in X$ uniformly at random, sample $W \sim \mu_{X-a}$, choose $T \subseteq B_a(X)$ according to $p(a, W, \cdot)$, and set $Z_p = W + \langle T \rangle$. To impose (15) on the distribution of Z_p, define, as in (3),

$$\begin{aligned} g_p(L) &= i\,\mathbb{E}[\dim(Z_p \cap L)] - (i-k)\dim(L) \\ &= -(i-k)\dim(L) + \sum_{a \in X} \sum_{W \in \mathrm{supp}(\mu_{X-a})} \mu_{X-a}(W) \sum_{T \subseteq B_a(X)} p(a, W, T)\dim((W + \langle T \rangle) \cap L). \end{aligned}$$

The expanded expression for $g_p(L)$ is affine in p. We impose the following constraints, where a ranges over X, W over $\mathrm{supp}(\mu_{X-a})$, and T over subsets of $B_a(X)$:

$$\begin{aligned} \mathrm{LP}_{\mathrm{XOS}}(X): \quad & p(a, W, T) \geq 0 && \forall\, a, W, T, && \text{(22a)} \\ & \sum_{T \subseteq B_a(X)} p(a, W, T) = 1 && \forall\, a, W, && \text{(22b)} \\ & p(a, W, T) = 0 && \text{if } \dim(W + \langle T \rangle) \neq \dim(W) + |T|, && \text{(22c)} \\ & \sum_{W} \mu_{X-a}(W) \sum_{\substack{T \subseteq B_a(X) \\ e \in T}} p(a, W, T) = \alpha && \forall\, a \in X,\ e \in B_a(X), && \text{(22d)} \\ & g_p(L) \leq 0 && \forall\, L \leq V_X. && \text{(22e)} \end{aligned}$$

Constraints (22a) and (22b) make $p(a, W, \cdot)$ a probability distribution. By (22c), every subset T assigned with positive probability satisfies $\dim(W + \langle T \rangle) = \dim(W) + |T|$. If the previously assigned elements are independent and span W, their union with T is independent, and T contains no previously assigned element. For each fixed $a \in X$, constraint (22d) imposes

$$\mathbb{P}[e \in T] = \sum_{W} \mu_{X-a}(W) \sum_{\substack{T \subseteq B_a(X) \\ e \in T}} p(a, W, T) = \alpha \qquad \text{for every } e \in B_a(X),$$

where the probability is over $W \sim \mu_{X-a}$ and the choice of T according to $p(a, W, \cdot)$. Finally, (22e) imposes

$$\mathbb{E}[\dim(Z_p \cap L)] \leq \left(1 - \frac{k}{i}\right)\dim(L) \qquad \text{for every } L \leq V_X,$$

where the expectation also includes the uniform choice of $a \in X$. Thus the distribution of Z_p satisfies (15).

Since the representation is over a finite field, V_X has finitely many subspaces and $\mathrm{LP}_{\mathrm{XOS}}(X)$ is a finite linear program. Over an arbitrary field, there are still only finitely many distinct dimension constraints, as explained in Remark 1. Each is determined by $\dim(L)$ and the dimensions of its intersections with the finitely many spans of subsets of $\bigcup_{a \in X} E_a$.

Let P_X be the polytope defined by (22a)–(22d). To solve $\mathrm{LP}_{\mathrm{XOS}}(X)$, we must find $p \in P_X$ such that $g_p(L) \leq 0$ for every $L \leq V_X$.

Lemma 7. *If μ_{X-a} satisfies* (15) *for every $a \in X$, then* $\mathrm{LP}_{\mathrm{XOS}}(X)$ *is feasible.*

Similar to Lemma 2, we prove Lemma 7 in Section 5.3 via an uncrossing argument. Suppose that each μ_{X-a} satisfies (15) and has rational entries. The linear program is feasible by Lemma 7, and its coefficients are rational, so it has a rational feasible solution. We compute such a solution from X and the data revealed by its agents, and denote it by p_X.

We use p_X to complete the recursive definition of μ_X. Recall that $Z_p = W + \langle T \rangle$, where a is uniform in X, $W \sim \mu_{X-a}$, and T is chosen according to $p(a, W, \cdot)$. Taking $p = p_X$, we set, for every $U \le V_X$,

$$\begin{aligned}\mu_X(U) &= \mathbb{P}[Z_{p_X} = U] \\ &= \frac{1}{i} \sum_{a \in X} \sum_{W \in \mathrm{supp}(\mu_{X-a})} \mu_{X-a}(W) \sum_{T \subseteq B_a(X)} p_X(a, W, T) \mathbf{1}[U = W + \langle T \rangle]. \end{aligned} \tag{23}$$

Constraint (22e) ensures that μ_X satisfies (15). Since p_X and the distributions μ_{X-a} have rational entries, (23) gives rational entries for μ_X. The base distributions $\mu_X = \delta_{\{0\}}$ for $|X| \le k$ have rational entries and satisfy (15) when $|X| = k$, so the recursion applies at every size.

XOS secretary algorithm. We store (μ_Y, p_Y) for subsets Y of the observed agents, together with $\mathrm{OPT}(Y)$ when $|Y| > k$. When a new agent a arrives, Algorithm 4 computes the entries for sets Y containing a, in increasing order of $|Y|$. For each $b \in Y$, the distribution μ_{Y-b} has then either been computed earlier in this update or stored before a arrived. Hence it is available when we solve $\mathrm{LP}_{\mathrm{XOS}}(Y)$. The deterministic choices of $\mathrm{OPT}(Y)$ and p_Y depend only on Y and its agents' data, so the stored entries do not depend on the arrival order.

Algorithm 3 calls Algorithm 4 before assigning a bundle to the current agent a. During the sample, it assigns the empty bundle. Otherwise, it samples $T_a \subseteq B_a(X)$ according to $p_X(a, \langle \overline{\mathrm{ALG}} \rangle, \cdot)$. When $n \ge 3$, we run $\mathcal{A}_k$ with $k = \lfloor n/e \rfloor \ge 1$. When $n \in \{1, 2\}$, we instead assign a maximum-value independent subset of E_a to the first arriving agent a and empty bundles to the remaining agents. In Algorithms 3 and 4, we obtain the vectors of E_a from the given representation or from agent a upon arrival, according to the model.

Algorithm 3. XOS Matroid Secretary ($\mathcal{A}_k$)

Input: Number of agents n and sample size $1 \le k < n$
$X \leftarrow \emptyset$; ALG $\leftarrow \emptyset$
foreach *arriving agent* a **do**
  Observe E_a, valuation f_a, and the vectors of E_a
  $X \leftarrow X + a$
  UpdateAllocationDistributions(X, a)
  $T_a \leftarrow \emptyset$
  if $|X| > k$ **then**
    Sample $T_a \subseteq B_a(X)$ according to $p_X(a, \langle \overline{\mathrm{ALG}} \rangle, \cdot)$
  Assign T_a irrevocably to a
  ALG $\leftarrow$ ALG $\cup \{(a, T_a)\}$
return ALG

For single-element arrivals, let each agent a_e have $E_{a_e} = \{e\}$ and $f_{a_e}(\{e\}) = w(e)$, and identify each observed agent set with the corresponding element set. Choose $\mathrm{OPT}(X)$ from the same greedy optimum as in Algorithm 1. Eliminating the empty-set variables from $\mathrm{LP}_{\mathrm{XOS}}(X)$ then gives $\mathrm{LP}(X)$, with $p_X(e, W) = p_X(a_e, W, \{e\})$ for $e \in \overline{\mathrm{OPT}(X)}$ and $p_X(e, W) = 0$ otherwise. If we choose the feasible solutions to correspond under this identification, Algorithms 3 and 4 recover Algorithms 1 and 2. In particular, after each arrival, $\overline{\mathrm{ALG}}$ has the same distribution as the accepted set in Algorithm 1.

Algorithm 4. UpdateAllocationDistributions(X, a)

Input: The observed agent set X, the newly arrived agent a, together with the revealed eligibility sets, valuations, and vectors of the agents in X

foreach $Y \subseteq X$ *with* $a \in Y$*, in increasing order of* $|Y|$ **do**
  if $|Y| \le k$ **then**
    Set $\mu_Y = \delta_{\{0\}}$ and $p_Y = 0$
  else
    Compute $\mathrm{OPT}(Y) = \{(b, B_b(Y)) : b \in Y\}$ using the deterministic tie-breaking rule for optimal allocations
    Enumerate the subspaces of V_Y
    Solve $\mathrm{LP}_{\mathrm{XOS}}(Y)$ using $(\mu_{Y-b})_{b \in Y}$, and compute a rational feasible solution p_Y deterministically from Y and its agents' data
    Compute μ_Y using (23) for the set Y
  Store (μ_Y, p_Y) and, when $|Y| > k$, $\mathrm{OPT}(Y)$

Lemma 8. *The algorithm $\mathcal{A}_k$ is well-defined and always produces a feasible allocation. Conditional on X being the set of the first $|X|$ arriving agents, the span $\left\langle \overline{\mathrm{ALG}} \right\rangle$ after processing X has distribution μ_X, which satisfies* (15) *when $|X| \ge k$. Moreover, for every $i > k$, every set X of i agents, every $a \in X$, and every $e \in B_a(X)$,*

$$\mathbb{P}[e \in T_a \mid X_i = X,\ a_i = a] = \frac{k}{i-1}, \tag{24}$$

where X_i is the set of the first i agents and a_i is the agent arriving in position i.

Proof. We use the same induction on $|X|$ as in Lemma 3. If $|X| \le k$, the algorithm assigns empty bundles, so the span is $\{0\}$ and has distribution $\mu_X = \delta_{\{0\}}$. For $|X| = k$, this distribution satisfies (15). Suppose that $|X| = i > k$. By induction, each μ_{X-a} satisfies (15). Lemma 7 therefore ensures that $\mathrm{LP}_{\mathrm{XOS}}(X)$ is feasible. Conditional on $X_i = X$ and $a_i = a$, the agents in $X - a$ arrive in uniformly random order. Thus the span W of the elements assigned before a arrives has distribution μ_{X-a}, and the algorithm chooses T_a according to $p_X(a, W, \cdot)$.

Conditional on $X_i = X$, the last agent a_i is uniform in X. Averaging over a_i, W, and T_a gives (23), so the span after processing X has distribution μ_X. Constraint (22e) gives (15), and constraint (22d) gives (24).

Finally, $T_a \subseteq B_a(X) \subseteq E_a$. By induction, the previously assigned elements are independent and span W. Constraint (22c) ensures that $\dim(W + \langle T_a \rangle) = \dim(W) + |T_a|$. Hence T_a contains no previously assigned element, and the union of all assigned bundles remains independent in M. □

5.3 Feasibility via Auxiliary Vectors and Uncrossing

Fix X with $|X| = i > k$ and distributions $(\mu_{X-a})_{a \in X}$ satisfying (15), and write $\alpha = k/(i-1)$. Applying Lemma 6 to each $B_a(X)$ with $W \sim \mu_{X-a}$ shows that P_X is nonempty.

As in the proof of Lemma 4, we first fix a chain of subspaces and find a point $p \in P_X$ satisfying its dimension constraints. We may choose a different p for each chain. Once we can do this for every chain, the same uncrossing argument gives a single $p \in P_X$ satisfying the constraints for all subspaces. In Section 3, we constructed the acceptance probabilities by ordering the predecessor subspaces W. Here, we add auxiliary vectors to each bundle $B_a(X)$ and apply Lemma 6 to the larger independent set.

Lemma 9. *Let $\alpha \in [0,1]$, and let W be a random subspace of V such that, for every $L \le V$,*

$$\mathbb{E}[\dim(W \cap L)] \le (1-\alpha)\dim(L).$$

Let B be an independent set of vectors in V, and put $H = \langle B \rangle$. For every chain $L_1 < \cdots < L_s \le V$, there is a randomized rule that, after observing W, selects $T \subseteq B$ satisfying $\dim(W + \langle T \rangle) = \dim(W) + |T|$ and $\mathbb{P}[b \in T] = \alpha$ for every $b \in B$. Moreover, for every $j \in \{1, \dots, s\}$,

$$\mathbb{E}[\dim((W + \langle T \rangle) \cap L_j)] \le (1-\alpha)\dim(L_j) + \alpha \dim(H \cap L_j).$$

Proof. To bound $\dim((W+\langle T\rangle)\cap L_j)$, we add auxiliary vectors in each L_j and select them together with T. We require the full selected set to be independent modulo W. The selected auxiliary vectors in L_j then span a subspace with trivial intersection with $W + \langle T \rangle$. Thus $\dim((W + \langle T \rangle) \cap L_j)$ is at most $\dim(L_j)$ minus the number of selected auxiliary vectors in L_j.

Extend B to a basis of $H + L_1$ by adding vectors from L_1. Continue along the chain, extending the basis of $H + L_{j-1}$ to a basis of $H + L_j$ by adding vectors from L_j. This gives nested sets $D_1 \subseteq \cdots \subseteq D_s$ such that $D_j \subseteq L_j$ and $B \cup D_j$ is a basis of $H + L_j$. In particular,

$$|D_j| = \dim(L_j) - \dim(H \cap L_j).$$

The auxiliary vectors in D_s need not correspond to elements of E. Lemma 6 applies to any independent set of vectors in V, so we can apply it to $B \cup D_s$. Let $R \subseteq B \cup D_s$ be the selected subset. We retain only $T = R \cap B$ and put $W' = W + \langle T \rangle$.

Every $b \in B$ satisfies $\mathbb{P}[b \in T] = \alpha$. Lemma 6 also gives $\dim(W + \langle R \rangle) = \dim(W) + |R|$. Thus $\dim(W + \langle T \rangle) = \dim(W) + |T|$, and the selected auxiliary vectors in $R \cap D_j$ are linearly independent and satisfy $W' \cap \langle R \cap D_j \rangle = \{0\}$. Since $R \cap D_j \subseteq L_j$, the subspaces $W' \cap L_j$ and $\langle R \cap D_j \rangle$ form a direct sum inside L_j, so

$$\dim(W' \cap L_j) \le \dim(L_j) - |R \cap D_j|.$$

Each vector of D_j is selected with probability α, so $\mathbb{E}[|R \cap D_j|] = \alpha |D_j|$. Taking expectations gives

$$\mathbb{E}[\dim(W' \cap L_j)] \le \dim(L_j) - \alpha|D_j| = (1-\alpha)\dim(L_j) + \alpha \dim(H \cap L_j). \qquad \square$$

Proof of Lemma 7. Fix a chain $L_1 < \cdots < L_s \le V_X$. For each $a \in X$, apply Lemma 9 to $B_a(X)$ with $W \sim \mu_{X-a}$ and $\alpha = k/(i-1)$. Let $p(a, W, T)$ be the conditional probability that the selection rule assigns T, given W. The resulting probabilities satisfy (22a)–(22d), so $p \in P_X$.

Put $H_a = \langle B_a(X) \rangle$. Since $\mathrm{OPT}(X)$ is feasible, its bundles are pairwise disjoint and $\overline{\mathrm{OPT}(X)}$ is independent in M. Thus the subspaces H_a form a direct sum. For every $L \le V_X$, the subspaces $H_a \cap L$ also form a direct sum contained in L, so

$$\sum_{a \in X} \dim(H_a \cap L) \le \dim(L). \tag{25}$$

When $B_a(X) = \{e\}$, we have $\dim(H_a \cap L) = \mathbf{1}[e \in L]$. Thus (25) extends the bound on the number of optimal elements lying in L used at the end of Section 3. Averaging the bound of Lemma 9 over the uniformly chosen agent $a \in X$ and using (25), we obtain, for every L in the chain,

$$\begin{aligned}\mathbb{E}[\dim(Z_p \cap L)] &\le (1-\alpha)\dim(L) + \frac{\alpha}{i}\sum_{a \in X} \dim(H_a \cap L) \\ &\le \left(1 - \alpha + \frac{\alpha}{i}\right)\dim(L) = \left(1 - \frac{k}{i}\right)\dim(L),\end{aligned}$$

so $g_p(L) \le 0$ throughout the chain.

We now apply the proof of Lemma 4, with P_X in place of P and V_X in place of $\langle X \rangle$. Since the field is finite, V_X has finitely many subspaces. The polytope P_X is nonempty, compact, and convex, and $g_p(L)$ is affine in p. For each fixed p, the function g_p is supermodular: $L \mapsto \dim(Z_p \cap L)$ is supermodular by the calculation in Section 3, taking expectations preserves this property, and $\dim(L)$ is modular. We have constructed a point of P_X satisfying the dimension constraints for each fixed chain. The proof of Lemma 4 therefore gives $p \in P_X$ such that $g_p(L) \le 0$ for all $L \le V_X$. □

5.4 The $1/e$ Guarantees

Recall that X_i is the set of the first i agents and a_i is the agent arriving in position i. For XOS valuations, we first bound the expected value assigned at position i using $\mathrm{OPT}(X_i)$. We then compare $F(\mathrm{OPT}(X_i))$ with the value obtained by restricting the fixed allocation $\mathrm{OPT}(N)$ to X_i.

Proof of Theorem 2. Suppose first that a finite-field representation is supplied and $n \ge 3$. Run $\mathcal{A}_k$ with $k = \lfloor n/e \rfloor$. Fix $i > k$ and a set X of i agents. Let $R_i = f_{a_i}(T_{a_i})$ be the value assigned at position i. Condition on $X_i = X$ and $a_i = a$. The bundle $B_a(X)$ is now fixed, and the algorithm assigns each of its elements with probability $k/(i-1)$ by (24). Since f_a is XOS, we apply (17) under this conditioning to obtain $\mathbb{E}[f_a(T_a) \mid X_i = X, a_i = a] \ge \frac{k}{i-1} f_a(B_a(X))$.

Conditional on $X_i = X$, the last agent a_i is uniform in X. Averaging over a gives

$$\begin{aligned}\mathbb{E}[R_i \mid X_i = X] &= \frac{1}{i} \sum_{a \in X} \mathbb{E}[f_a(T_a) \mid X_i = X,\ a_i = a] \\ &\ge \frac{k}{i(i-1)} \sum_{a \in X} f_a(B_a(X)) = \frac{k}{i-1} \frac{F(\mathrm{OPT}(X))}{i}. \end{aligned} \tag{26}$$

Write $\mathrm{OPT}(N) = \{(a, B_a^*) : a \in N\}$ for the fixed optimal allocation for all agents. Restricting $\mathrm{OPT}(N)$ to the agents of X_i yields a feasible allocation for X_i, and each agent $a \in N$ belongs to X_i with probability i/n, so

$$\mathbb{E}[F(\mathrm{OPT}(X_i))] \ge \mathbb{E}\left[\sum_{a \in X_i} f_a(B_a^*)\right] = \frac{i}{n} F(\mathrm{OPT}(N)). \tag{27}$$

Taking expectations in (26), using (27), and summing over $i \in \{k+1, \dots, n\}$ gives

$$\mathbb{E}[F(\mathrm{ALG})] \ge \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j} F(\mathrm{OPT}(N)) \ge \frac{1}{e} F(\mathrm{OPT}(N)).$$

The last inequality is the harmonic-sum bound proved after (5) in Section 2.3 for $k = \lfloor n/e \rfloor$.

For $n \in \{1, 2\}$, assigning a maximum-value independent subset of E_a to the first arriving agent a and empty bundles to the remaining agents gives

$$\mathbb{E}[F(\mathrm{ALG})] \ge \frac{1}{n} \sum_{a \in N} f_a(B_a^*) = \frac{1}{n} F(\mathrm{OPT}(N)) \ge \frac{1}{e} F(\mathrm{OPT}(N)).$$

In the known-matroid model, if no finite-field representation is supplied, we compute one before the arrivals by the enumeration argument in the proof of Corollary 1 and run the same algorithm. The same algorithm also works when a representation over a known finite field is revealed with the agents, in a common coordinate system. For every observed set Y, we compute $\mathrm{OPT}(Y)$ and V_Y from the eligibility sets, valuations, and vectors revealed by the agents in Y. All computations in Algorithm 4 therefore use information available at that point. □

For group arrivals, we fix an element of the global greedy optimum. By the argument in Section 2.3, this element belongs to the greedy optimum of every prefix containing its group. We then bound its probability of acceptance.

Proof of Corollary 3. Let $I^* = \overline{\mathrm{OPT}(N)} = I(N)$ be the global greedy optimum under the fixed tie-breaking order. For every $e \in I^*$ and every set of groups X containing its group, we have $e \in I(X)$. Indeed, e is not spanned by the elements preceding it in the global greedy order. Restricting those elements to $\bigcup_{a \in X} G_a$ cannot make them span e.

Suppose $n \geq 3$ and run $\mathcal{A}_k$ with $k = \lfloor n/e \rfloor$ and bundles $B_a(X) = I(X) \cap G_a$. Fix $e \in I^*$ and let a be its group. Conditional on $X_i = X$ and $a_i = a$ for $i > k$, we have $e \in B_a(X)$, so Lemma 8 gives $\mathbb{P}[e \in T_a \mid X_i = X,\ a_i = a] = k/(i-1)$. This probability is the same for every possible X, so $\mathbb{P}[e \in T_a \mid a_i = a] = k/(i-1)$. Averaging over the uniformly distributed arrival position of group a gives

$$\mathbb{P}[e \in \overline{\mathrm{ALG}}] = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j} \geq \frac{1}{e}.$$

For $n \in \{1, 2\}$, accepting the greedy independent set of the first arriving group and rejecting the remaining groups accepts each $e \in I^*$ whenever its group arrives first, with probability $1/n \geq 1/e$.

In the online-representation model, for every observed subset Y, both the greedy optimum $I(Y)$ and the ambient subspace $V_Y = \langle \bigcup_{a \in Y} G_a \rangle$ used in Algorithm 4 depend only on the groups in Y and their revealed vectors and weights. Thus Algorithms 3 and 4 can be executed online without prior knowledge of the full matroid, with the same acceptance probabilities. □

Proof of Corollary 4. In the unit-demand formulation, we chose each $\mathrm{OPT}(X)$ so that $|B_a(X)| \leq 1$ for every $a \in X$. Since $\mathcal{A}_k$ assigns only subsets of $B_a(X)$, every agent receives at most one element, and Theorem 2 gives the $1/e$ expected-value guarantee (with the same singleton restriction on the first agent's bundle when $n \in \{1, 2\}$). □

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AI Methodology.

We had been working for some time on an approach to the Matroid Secretary Problem that assigns correlated acceptance probabilities equal to $k/(i-1)$ in order to achieve a $1/e$ competitive ratio, while maintaining a subspace dimension invariant. Motivated by Buchbinder, Jain, and Singh [11], we formulated this step by a linear program, focusing primarily on binary matroids. While trying to prove the feasibility of this LP, we discussed the problem with ChatGPT-6 Astra, which identified the supermodularity of g_p and proposed the uncrossing argument used to establish feasibility. The

resulting proof, which forms the basis of Section 3, was initially drafted with assistance from ChatGPT-6 Astra. Building on this feasibility argument, we developed the extensions in Sections 4 and 5. All arguments and references were checked by the authors, who are responsible for their verification and final presentation.

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